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Lin et al.

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(54) **METHOD TO REDUCE BREAKDOWN FAILURE IN A MIM CAPACITOR**

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(2013.01); **H01L 21/02315** (2013.01);
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CPC H01L 21/7685; H01L 21/76822; H10D 1/696

See application file for complete search history.

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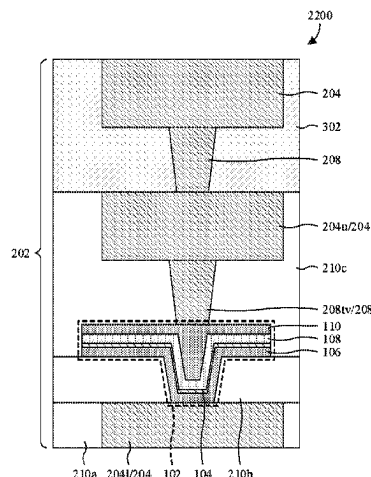
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(57) **ABSTRACT**

Various embodiments of the present application are directed towards a method for forming a metal-insulator-metal (MIM) capacitor comprising an enhanced interfacial layer to reduce breakdown failure. In some embodiments, a bottom electrode layer is deposited over a substrate. A native oxide layer is formed on a top surface of the bottom electrode layer and has a first adhesion strength with the top surface. A plasma treatment process is performed to replace the native oxide layer with an interfacial layer. The interfacial layer is conductive and has a second adhesion strength with the top surface of the bottom electrode layer, and the second adhesion strength is greater than the first adhesion strength. An insulator layer is deposited on the interfacial layer. A top

(Continued)



electrode layer is deposited on the insulator layer. The top and bottom electrode layers, the insulator layer, and the interfacial layer are patterned to form a MIM capacitor.

20 Claims, 19 Drawing Sheets

Related U.S. Application Data

division of application No. 16/579,738, filed on Sep. 23, 2019, now Pat. No. 11,152,455.

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H10B 61/00 (2023.01)

H10D 1/68 (2025.01)

(52) U.S. Cl.

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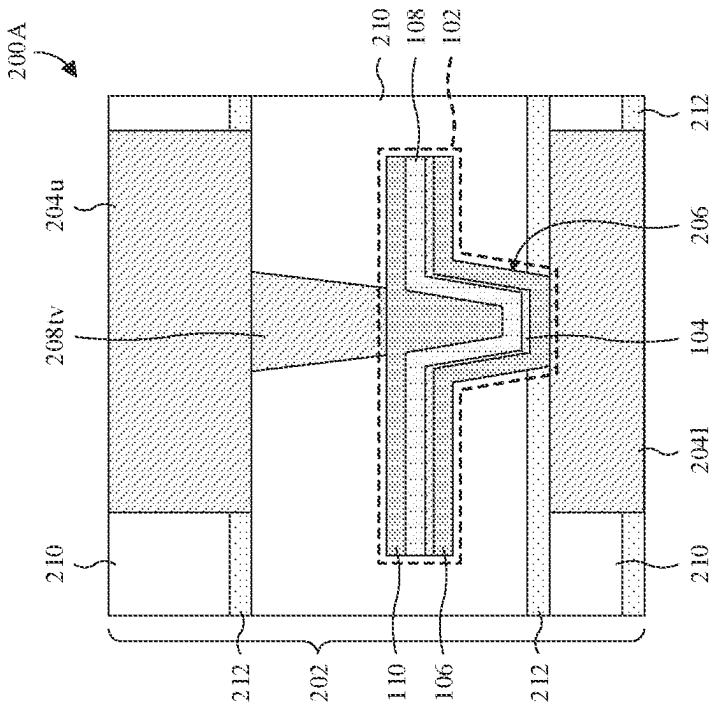


Fig. 2A

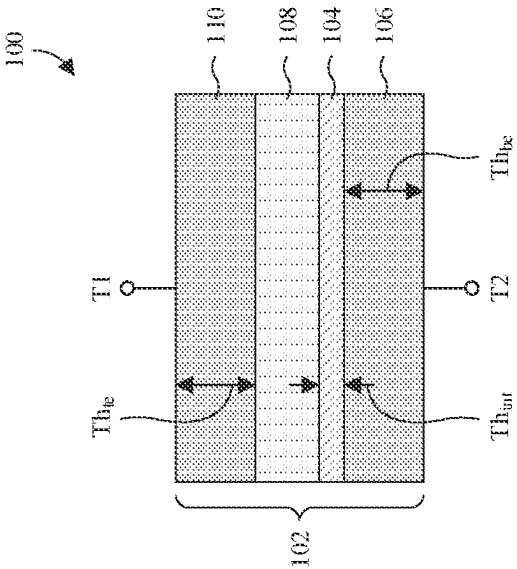


Fig. 1

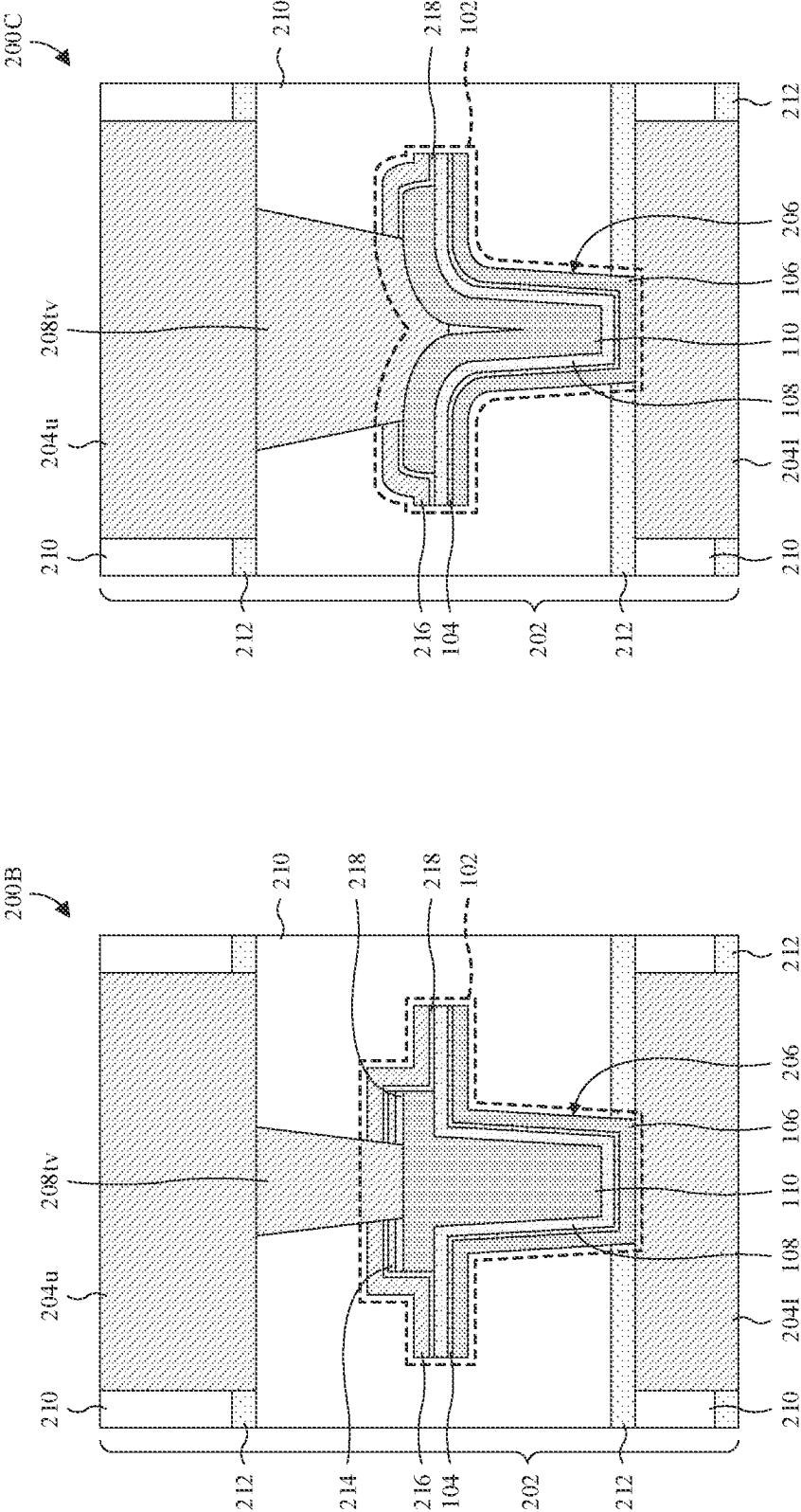


Fig. 2C

Fig. 2B

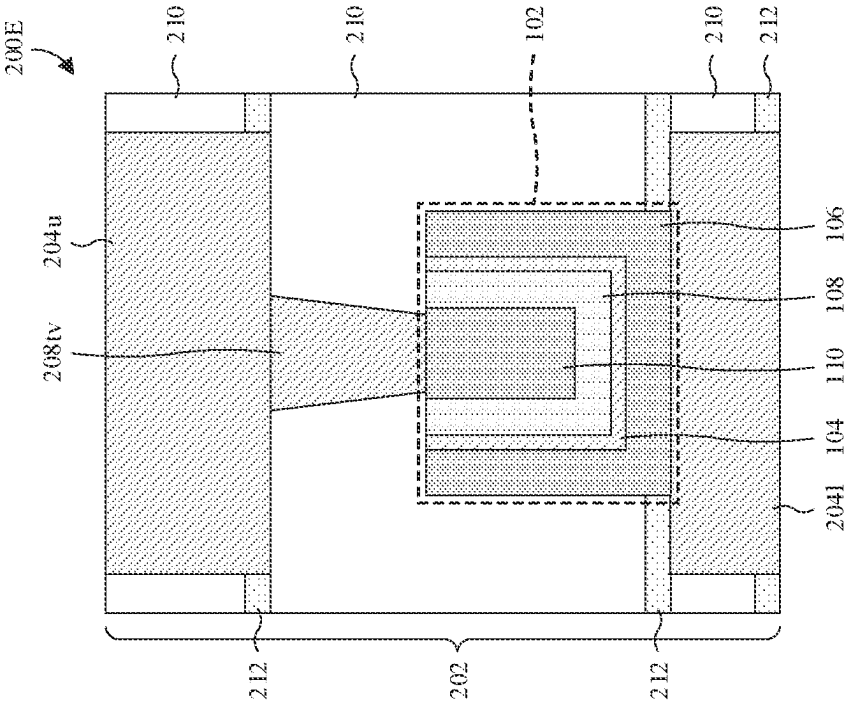


Fig. 2E

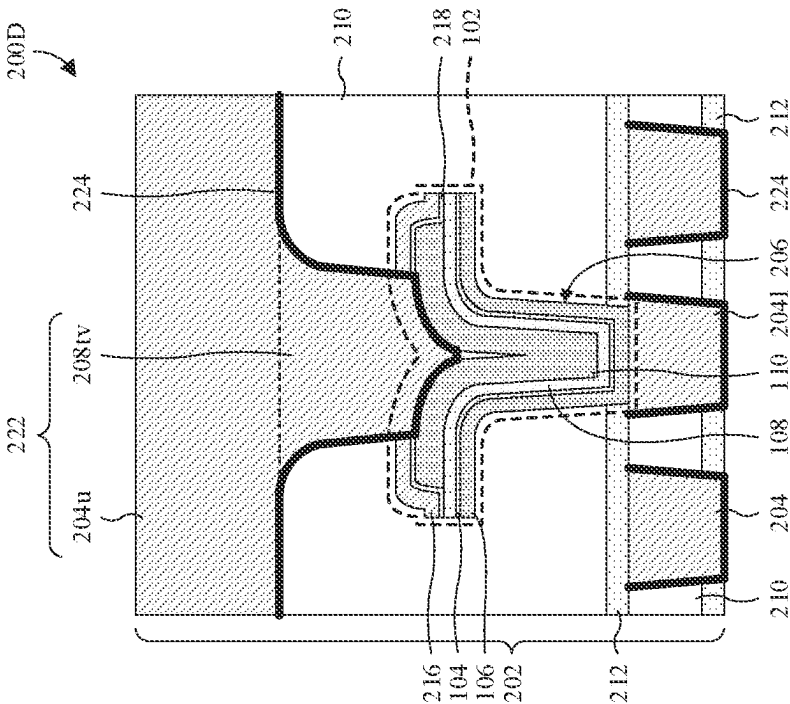


Fig. 2D

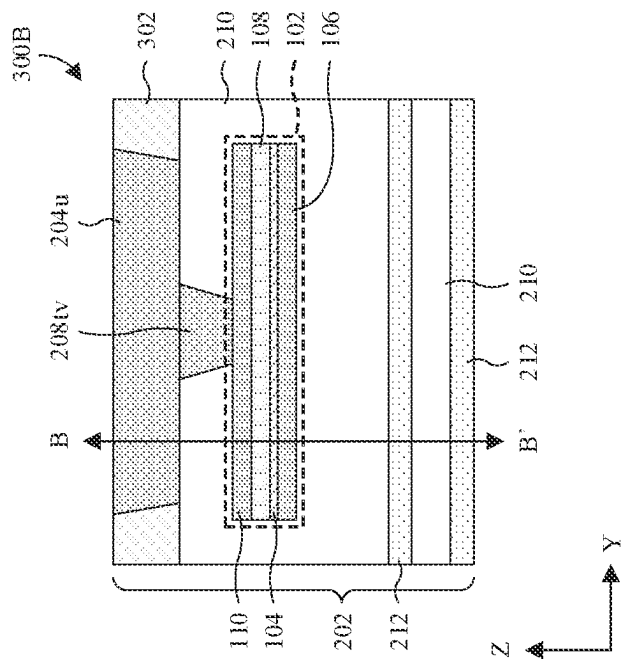


Fig. 3B

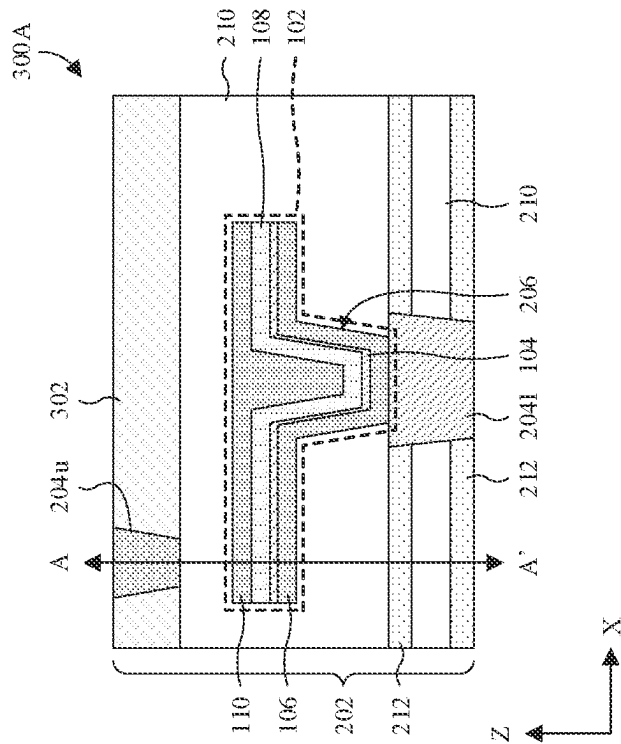


Fig. 3A

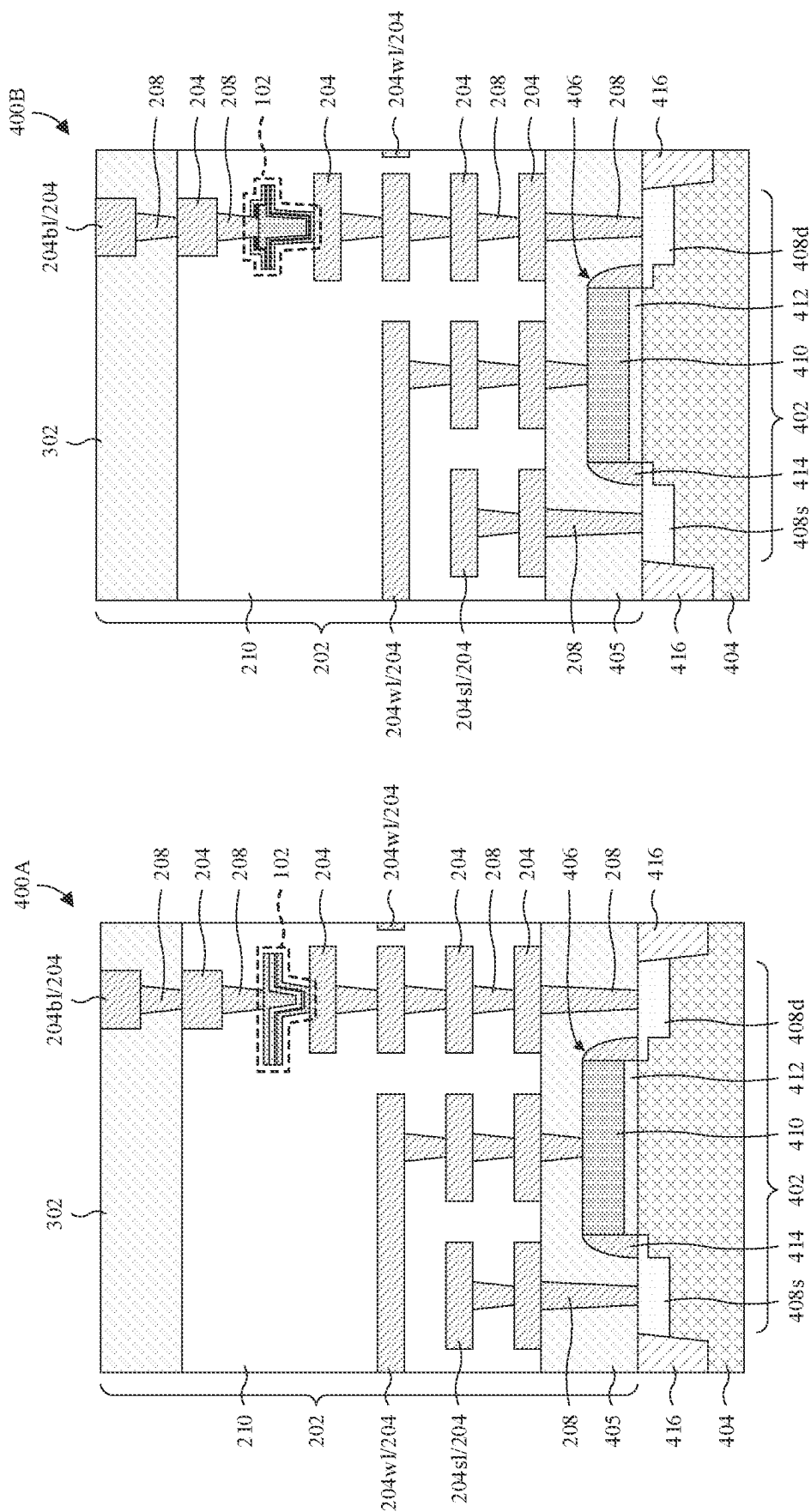


Fig. 4B

Fig. 4A

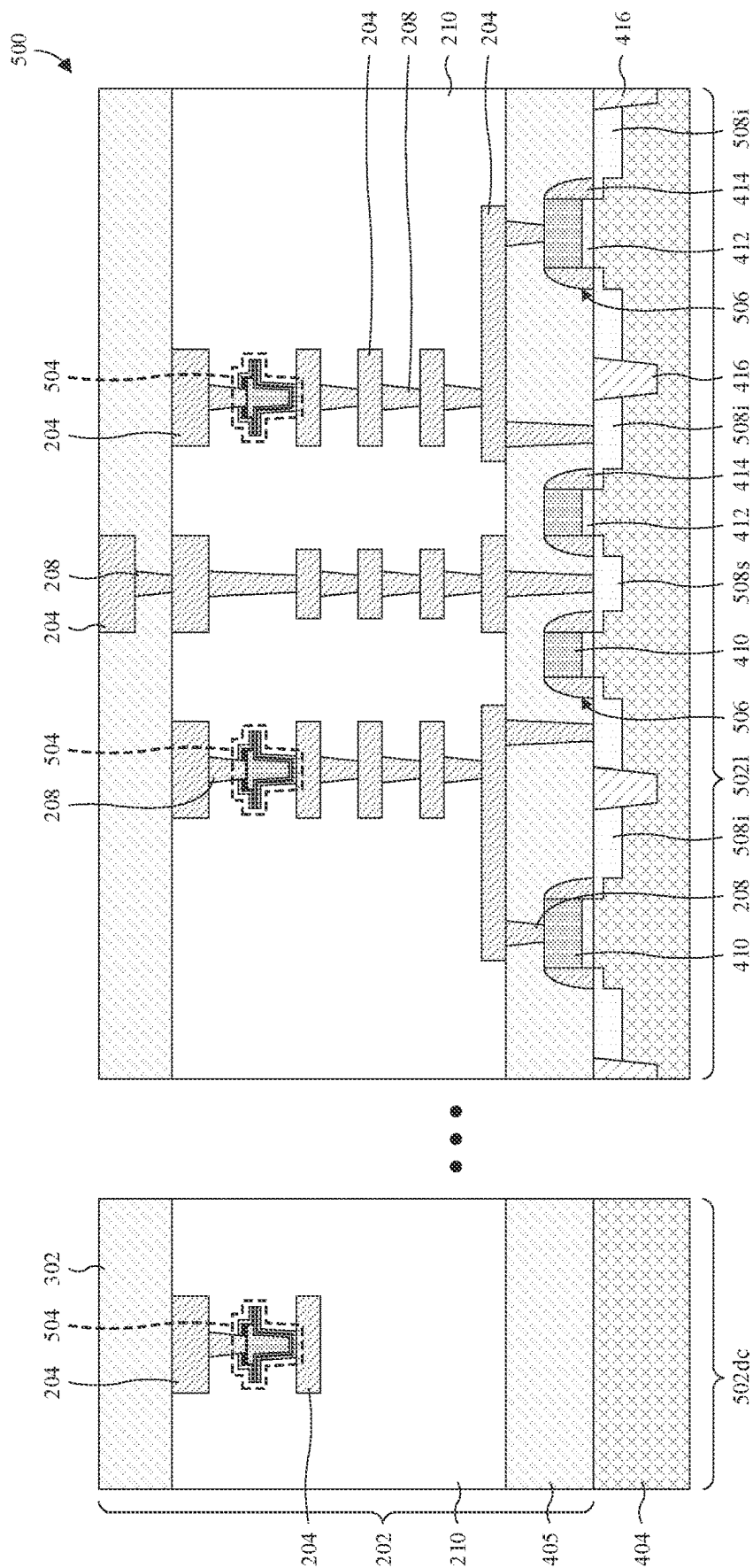


Fig. 5

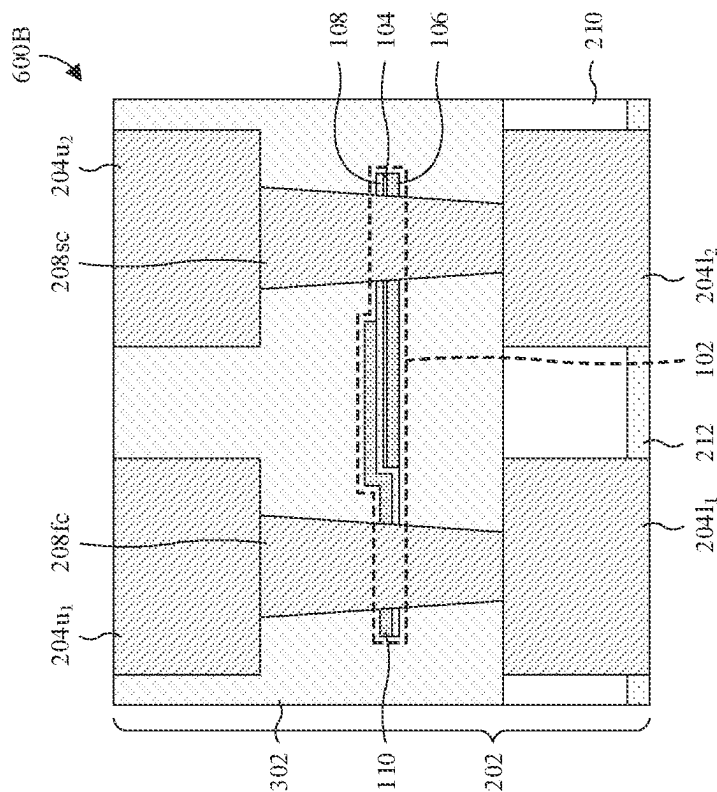


Fig. 6B

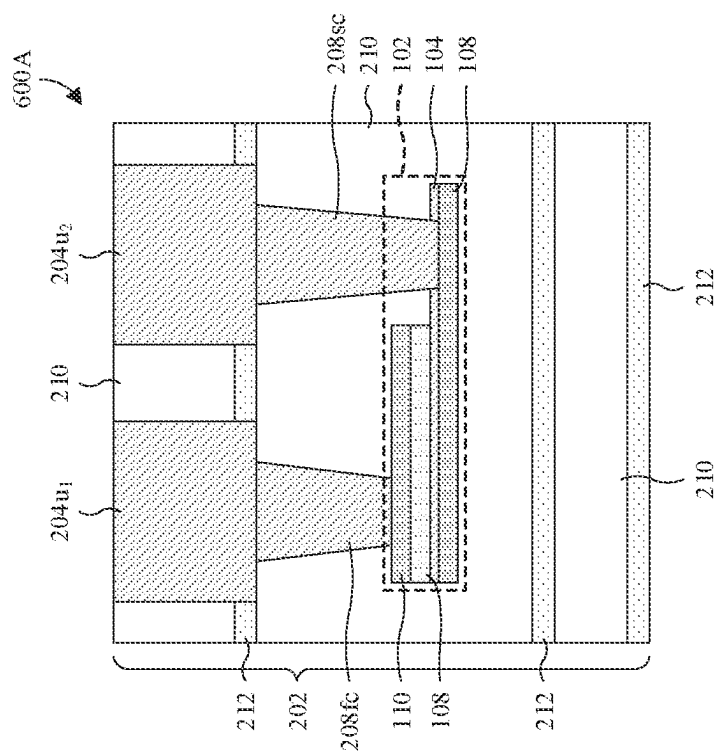


Fig. 6A

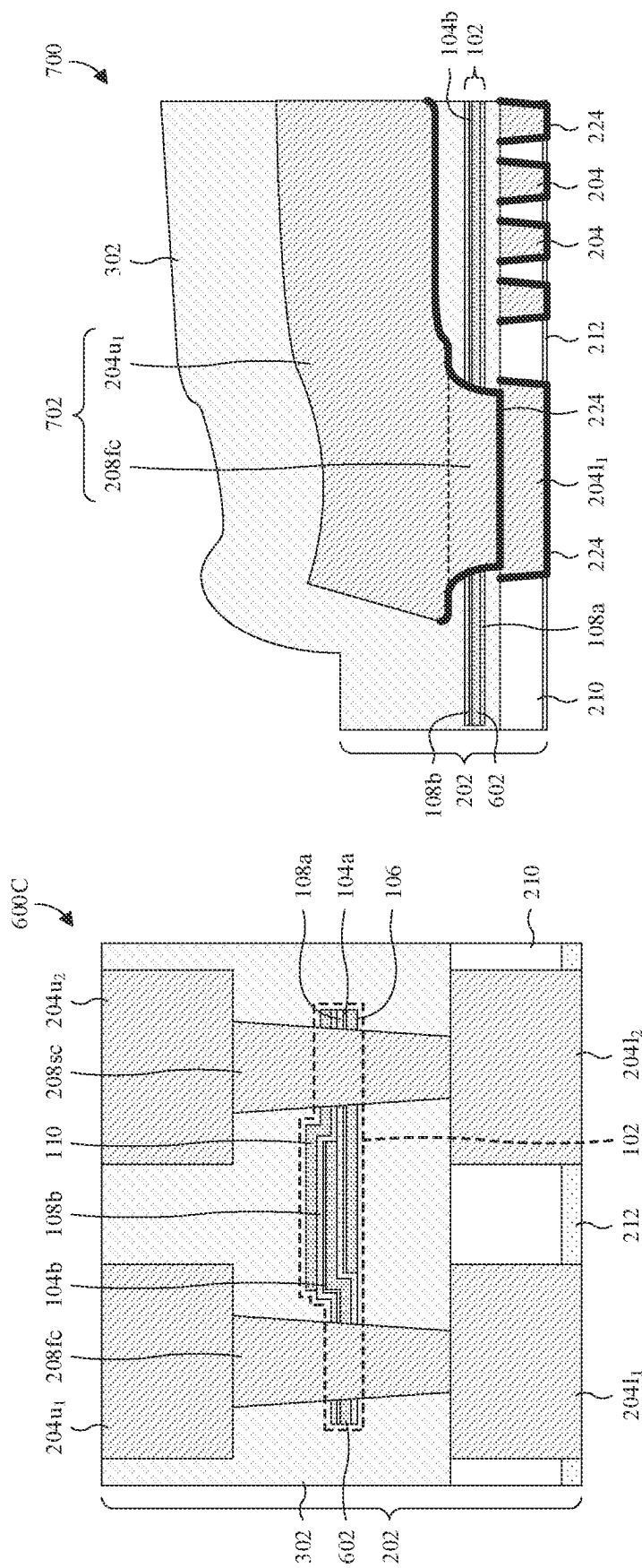
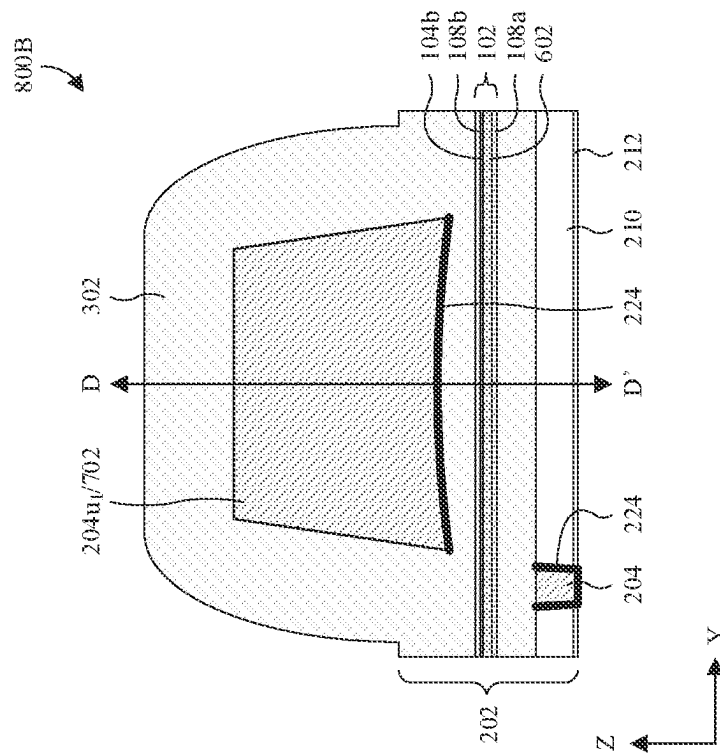


Fig. 7

Fig. 6C



Fi. 8.

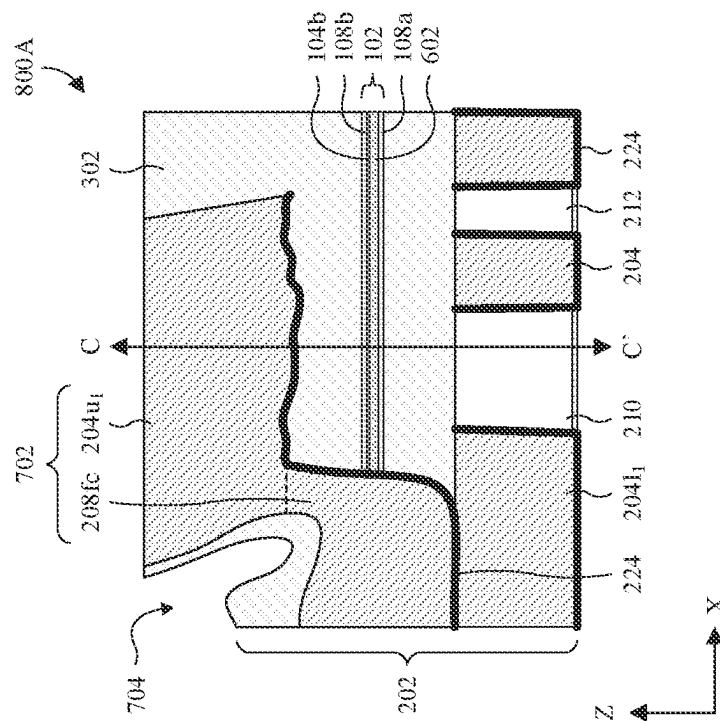


Fig. 8A

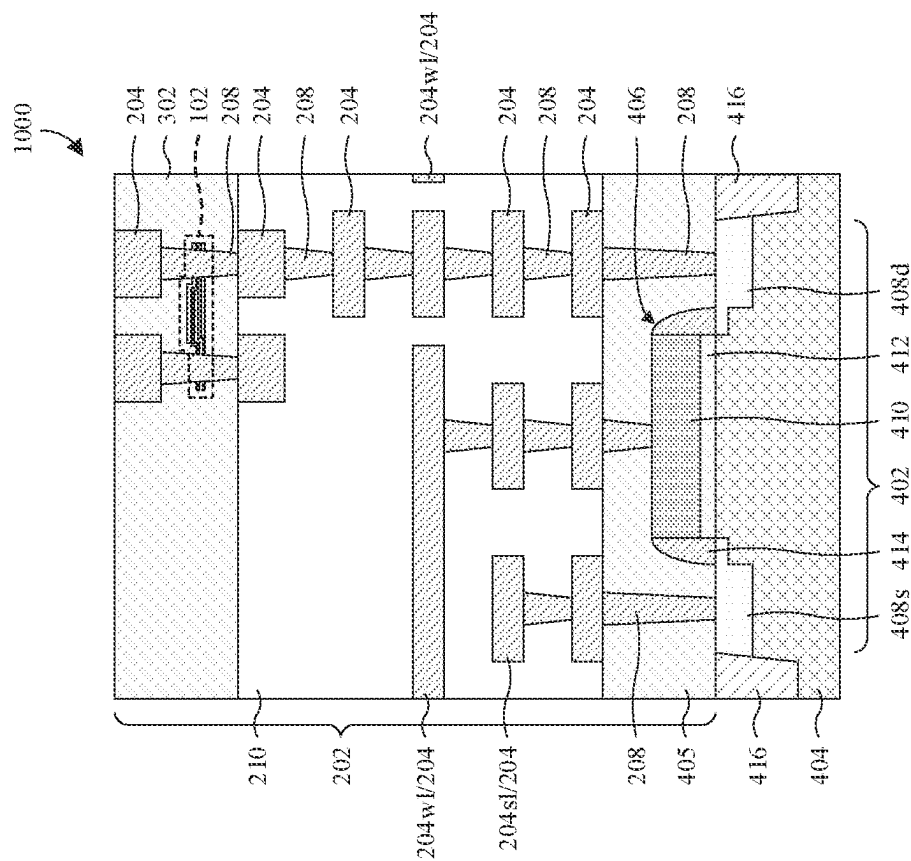
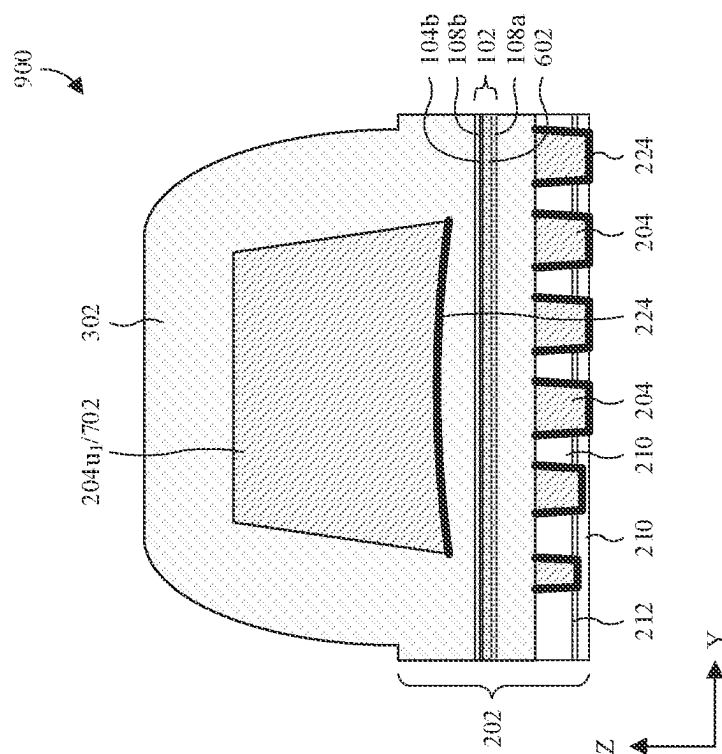


Exhibit 10



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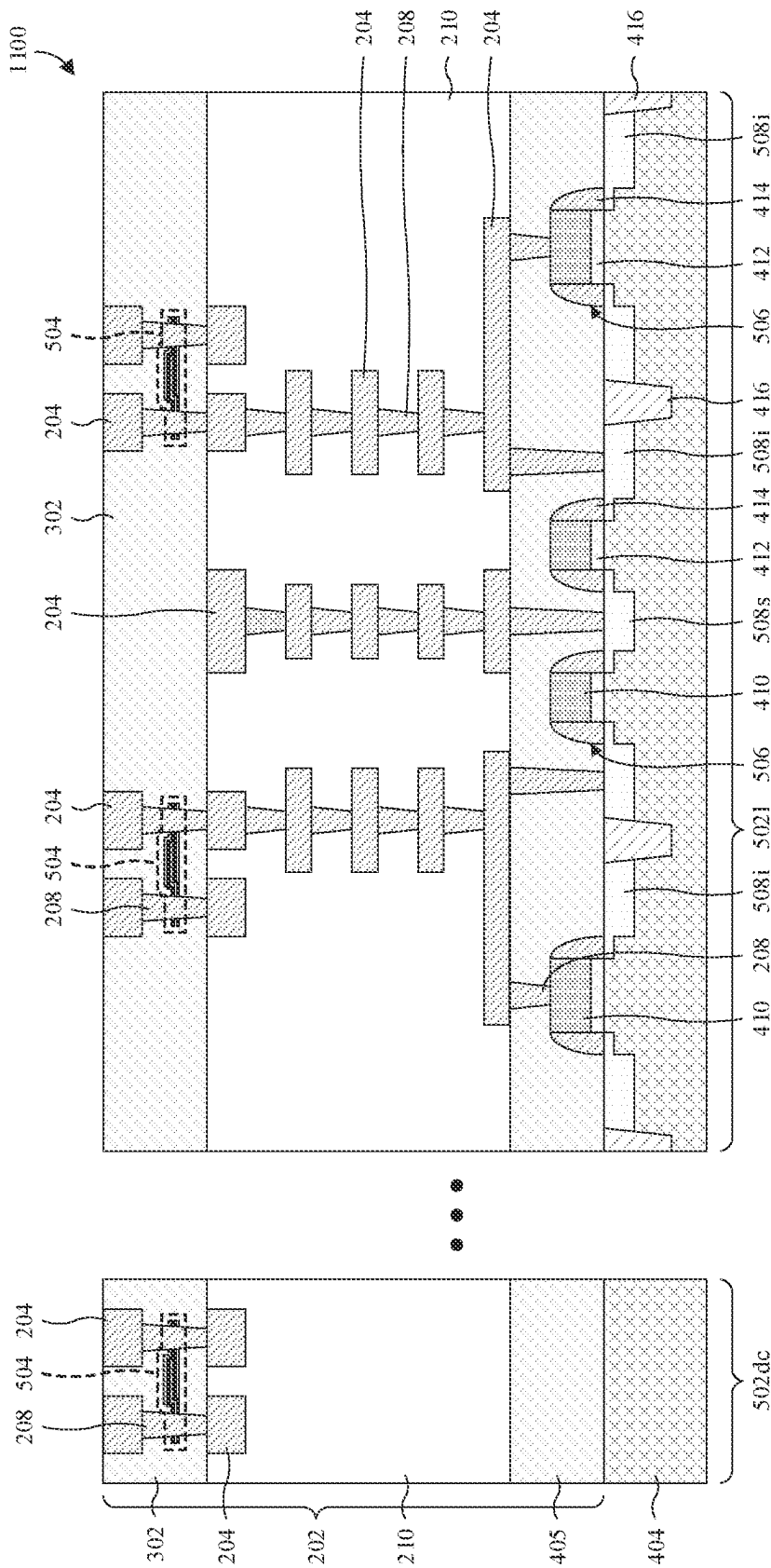


Fig. 11

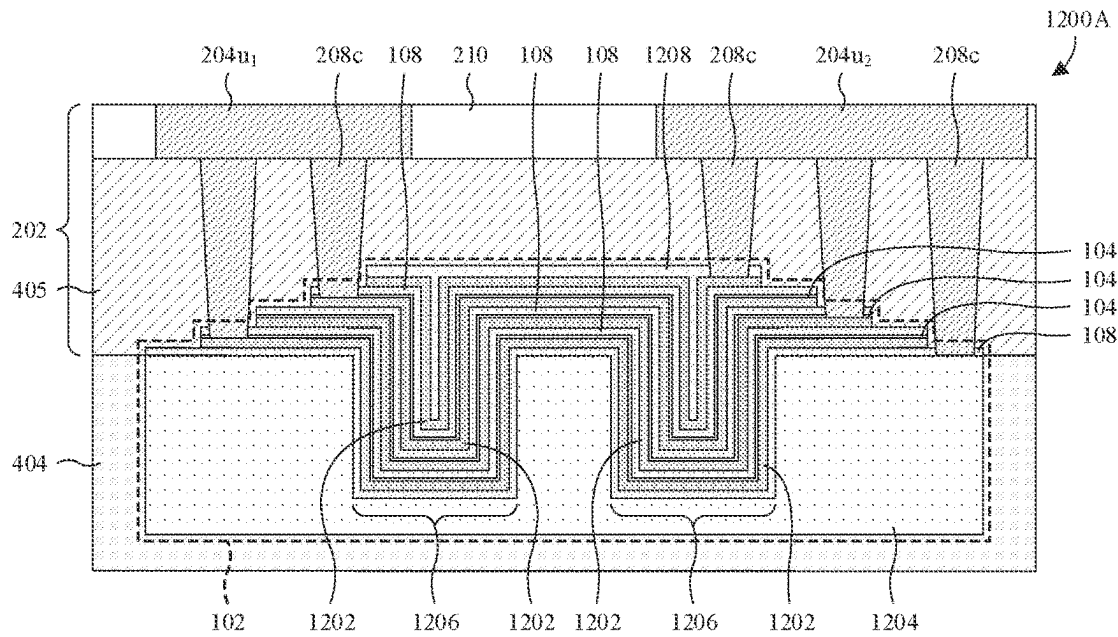


Fig. 12A

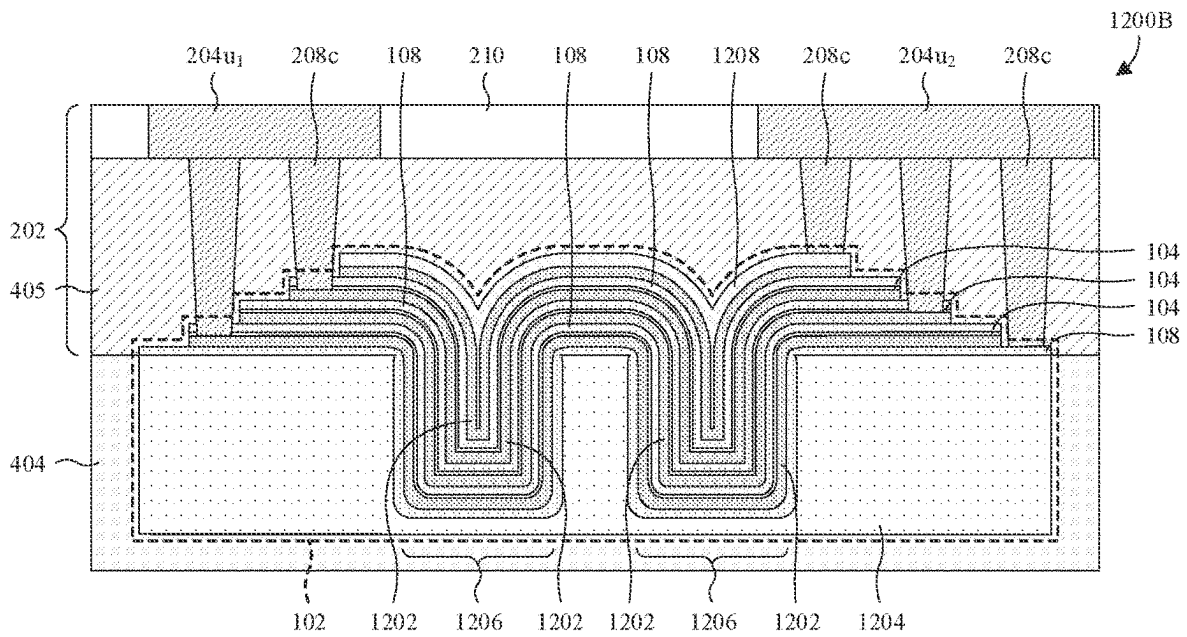
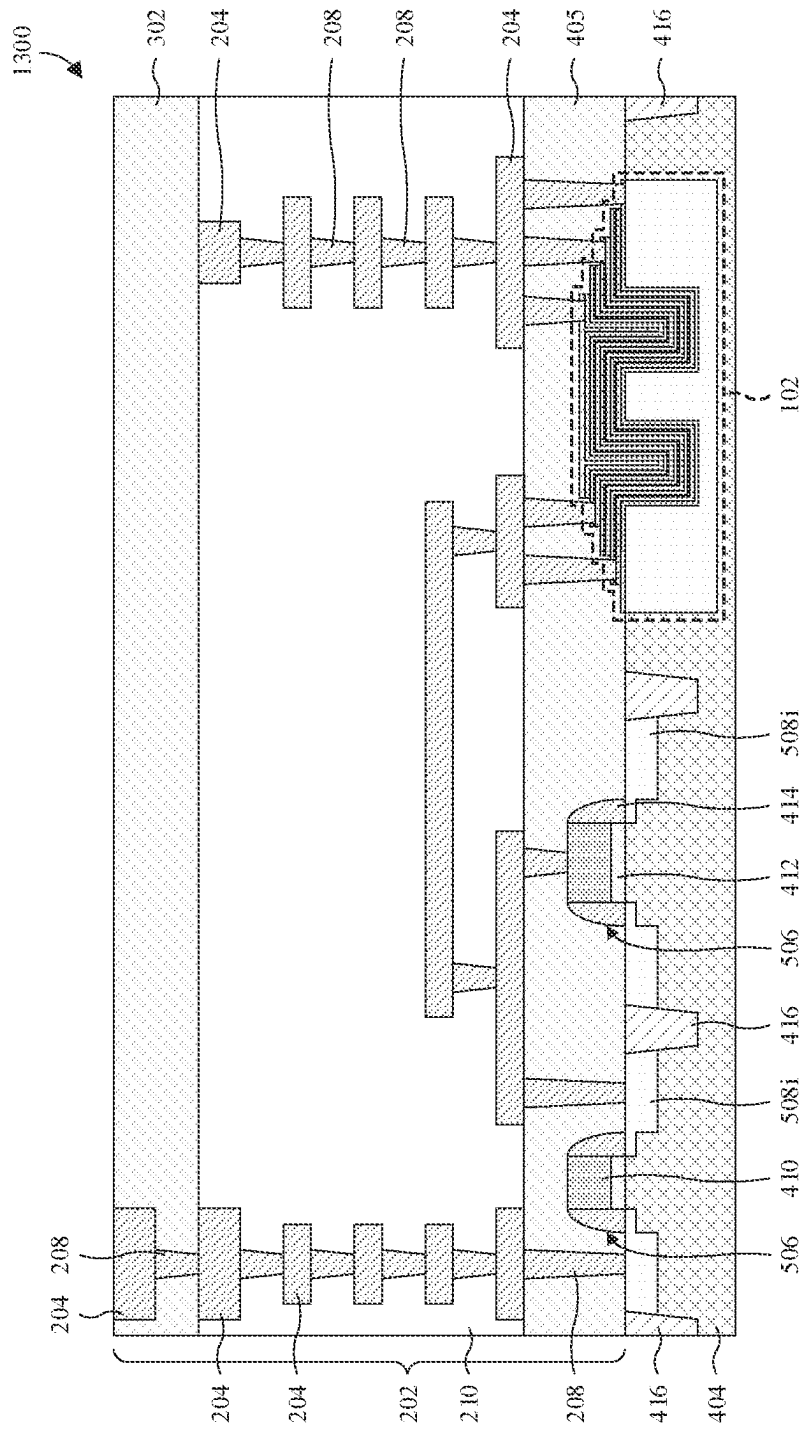


Fig. 12B

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Lib

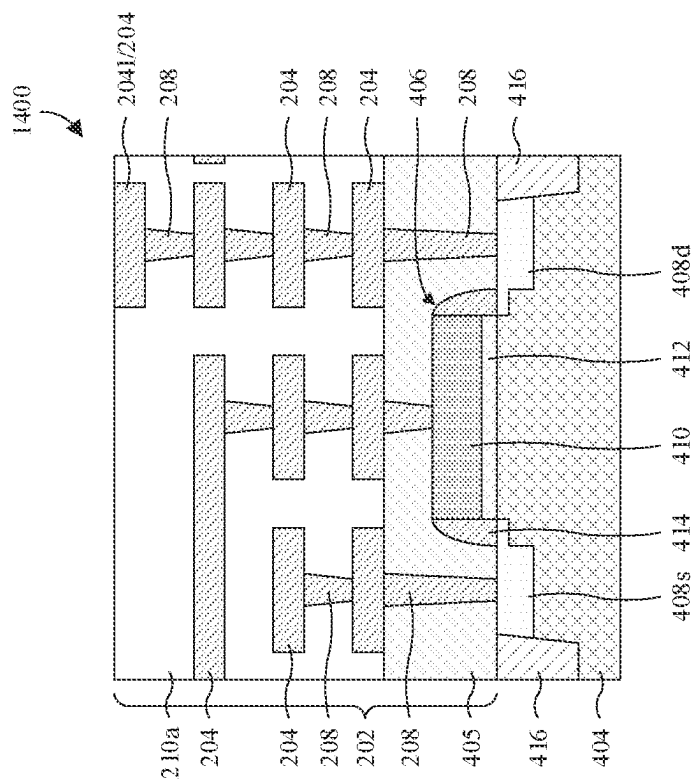


Fig. 14

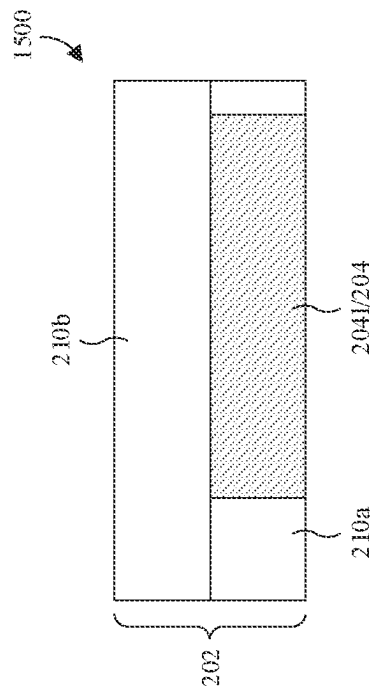


Fig. 15

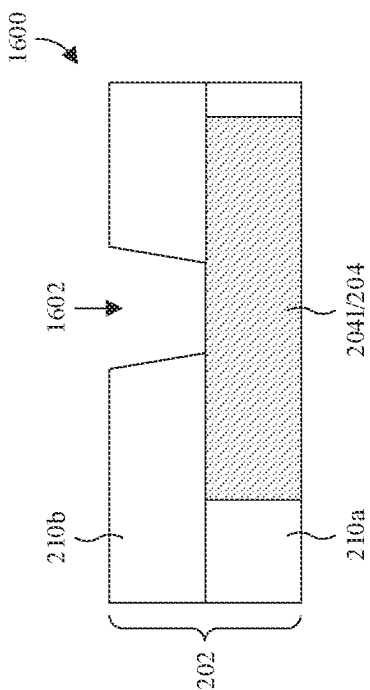


Fig. 16

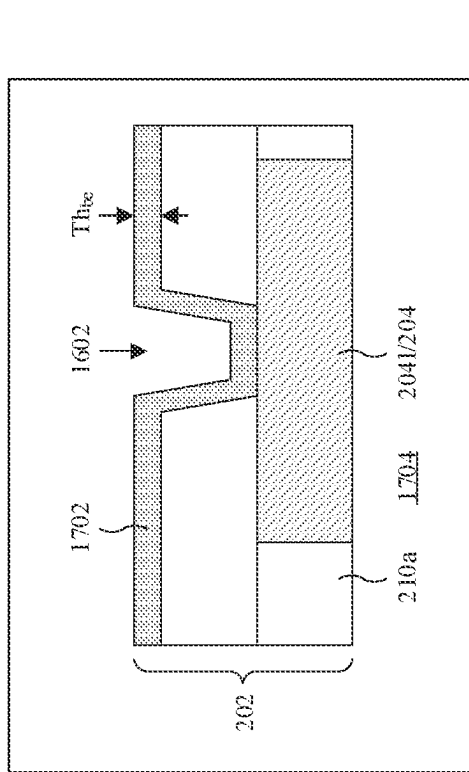


Fig. 17

1800

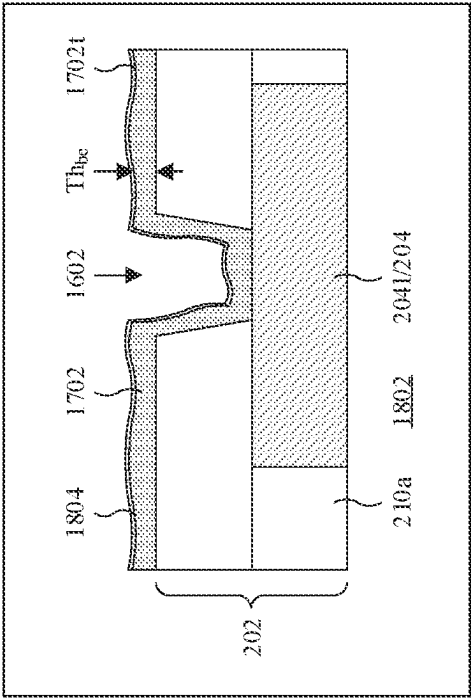


Fig. 18

1900

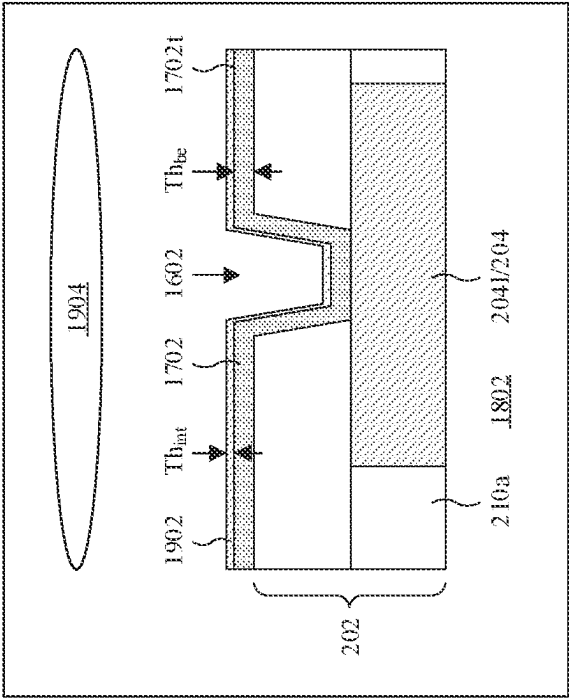


Fig. 19

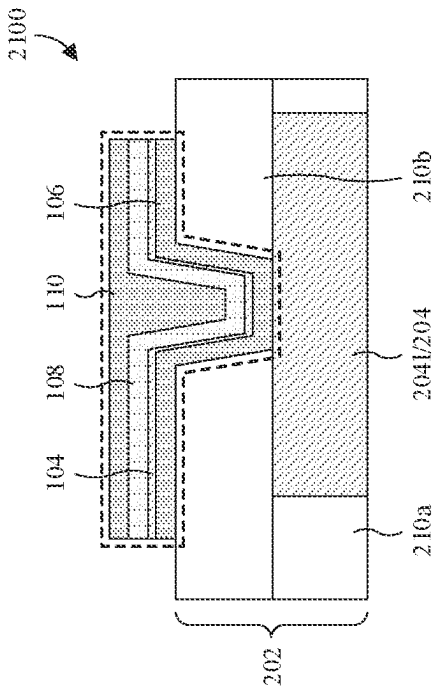


Fig. 21

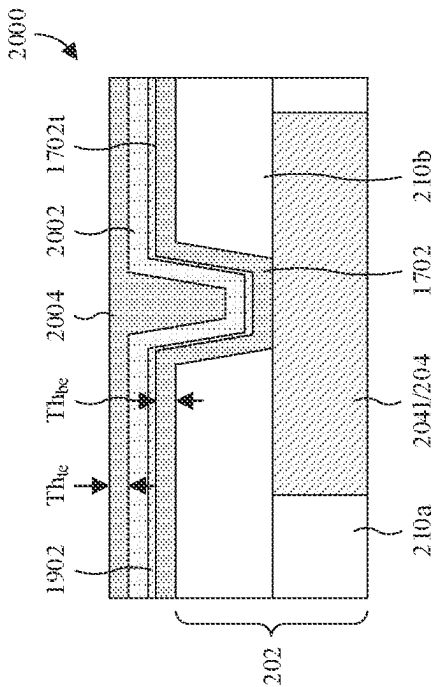
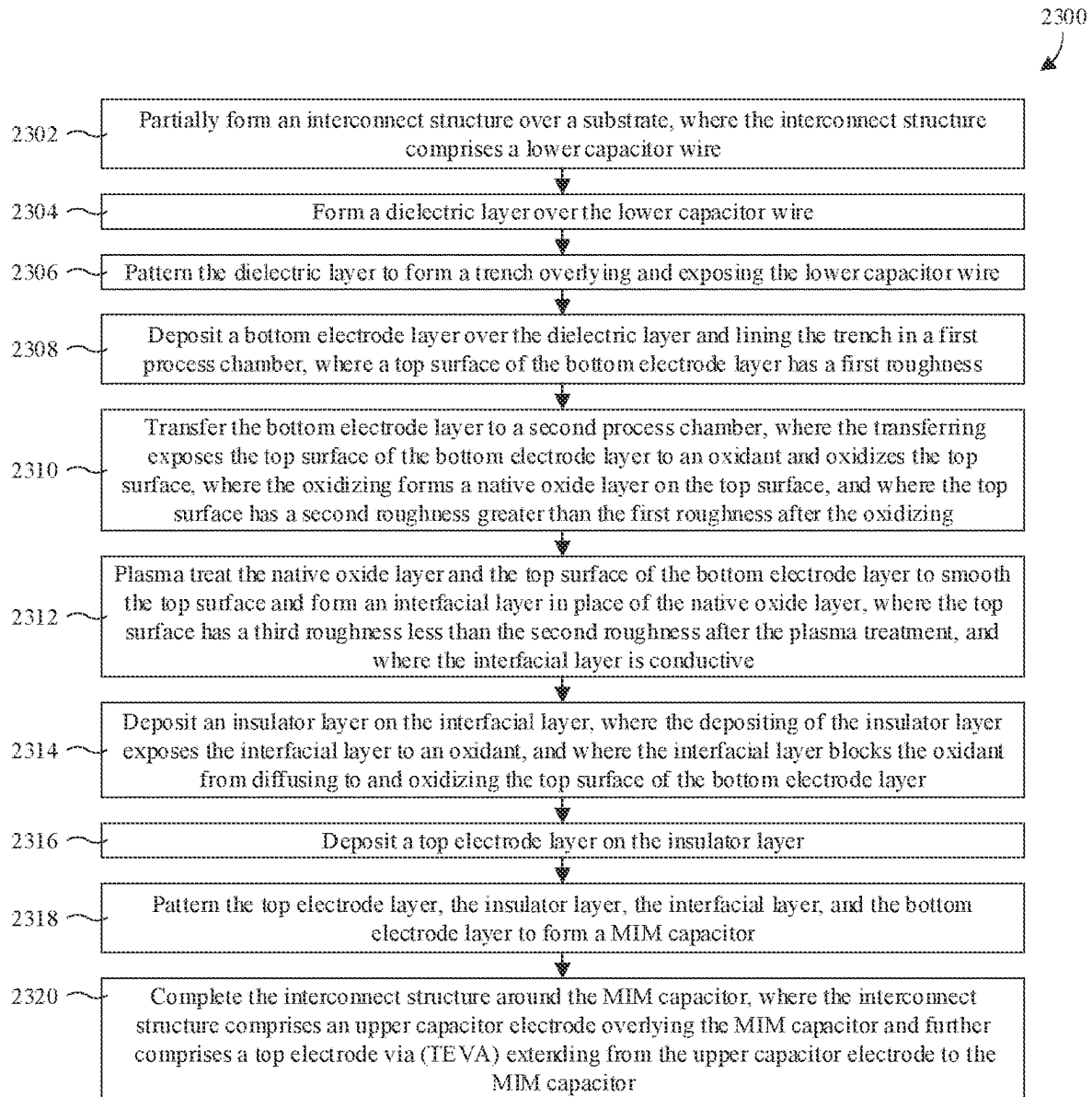


Fig. 20

**Fig. 23**

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METHOD TO REDUCE BREAKDOWN FAILURE IN A MIM CAPACITOR

REFERENCE TO RELATED APPLICATIONS

This Application is a Continuation of U.S. application Ser. No. 17/501,269, filed on Oct. 14, 2021, which is a Divisional of U.S. application Ser. No. 16/579,738, filed on Sep. 23, 2019 (now U.S. Pat. No. 11,152,455, issued on Oct. 19, 2021). The contents of the above-referenced Patent Applications are hereby incorporated by reference in their entirety.

BACKGROUND

Integrated circuits (ICs) are formed on semiconductor dies comprising millions or billions of transistor devices. The transistor devices are configured to act as switches and/or to produce power gains so as to enable logical functionality. ICs also comprise passive devices used to control gains, time constants, and other IC characteristics. One type of passive device is a metal-insulator-metal (MIM) capacitor.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 illustrates a cross-sectional view of some embodiments of a metal-insulator-metal (MIM) capacitor comprising an interfacial layer enhanced to reduce breakdown failure.

FIGS. 2A-2E illustrate cross-sectional views of some embodiments of an integrated circuit (IC) chip comprising an interconnect structure in which some first trench embodiments of the MIM capacitor of FIG. 1 are embedded.

FIGS. 3A and 3B illustrate orthogonal cross-sectional views of some alternative embodiments of the IC chip of FIG. 2A in which an upper capacitor wire is in a passivation layer.

FIGS. 4A and 4B illustrate cross-sectional views of some embodiments of an IC chip comprising a one-transistor one-capacitor (1T1C) cell in which the MIM capacitors of FIGS. 2A and 2B are embedded.

FIG. 5 illustrates a cross-sectional view of some embodiments of an IC chip comprising a decoupling-capacitor region and a logic region in which MIM capacitors are at the decoupling-capacitor and logic regions and configured as the MIM capacitor of FIG. 2B.

FIGS. 6A-6C illustrate cross-sectional views of some embodiments of an IC chip comprising an interconnect structure in which planar embodiments of the MIM capacitor of FIG. 1 are embedded.

FIG. 7 illustrates a cross-sectional view of some alternative embodiments of a portion of the IC chip of FIG. 6C at a first capacitor via.

FIGS. 8A and 8B illustrate orthogonal cross-sectional views of some alternative embodiments of a portion of the IC chip of FIG. 7.

FIG. 9 illustrates a cross-sectional view of some alternative embodiments of the IC chip of FIG. 8B.

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FIG. 10 illustrates a cross-sectional view of some embodiments of an IC chip comprising a 1T1C cell in which the MIM capacitor of FIG. 6C is embedded.

FIG. 11 illustrates a cross-sectional view of some embodiments of an IC chip comprising a decoupling-capacitor region and a logic region in which MIM capacitors are at the decoupling-capacitor and logic regions and configured as the MIM capacitor of FIG. 6C.

FIGS. 12A and 12B illustrates cross-sectional views of some embodiments of an IC chip comprising a substrate in which second trench embodiments of the MIM capacitor of FIG. 1 are embedded.

FIG. 13 illustrates an expanded cross-sectional view of some embodiments of the IC chip of FIG. 12A.

FIGS. 14-22 illustrate a series of cross-sectional views of some embodiments of a method for forming an IC chip in which a MIM capacitor comprises an interfacial layer enhanced to reduce breakdown failure.

FIG. 23 illustrates a block diagram of some embodiments of the method of FIGS. 14-22.

DETAILED DESCRIPTION

The present disclosure provides many different embodiments, or examples, for implementing different features of this disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

A metal-insulator-metal (MIM) capacitor comprises a bottom electrode, a capacitor insulator layer overlying the bottom electrode, and a top electrode overlying the capacitor insulator layer. A method for forming the MIM capacitor may, for example, comprise depositing a bottom electrode layer over a substrate, depositing an insulator layer on the bottom electrode layer, depositing a top electrode layer on the insulator layer, and patterning the various layers (e.g., the insulator layer) into the MIM capacitor. Challenges may, however, arise at least when the bottom electrode layer and the insulator layer are formed of titanium nitride and a high k dielectric. A high k dielectric may, for example, be a dielectric material having a dielectric constant greater than about 10 or some other suitable value.

When the bottom electrode layer and the insulator layer are formed of titanium nitride and a high k dielectric

material, the bottom electrode layer and the insulator layer may be formed in separate process chambers. For example, the bottom electrode layer may be formed in a physical vapor deposition (PVD) process chamber, whereas insulator layer may be formed in an atomic layer deposition (ALD) process chamber. However, forming the bottom electrode layer and the insulator layer in separate process chambers may lead to air exposure. The air exposure leads to oxidation of a top surface of the bottom electrode layer and hence to formation of a native oxide layer on the top surface. The oxidation increases a roughness of the top surface of the bottom electrode, which may degrade electric field uniformity across the capacitor insulator layer and may hence degrade performance of the MIM capacitor. The native oxide layer has weak adhesion with the top surface of the bottom electrode layer and is hence prone to delamination from the top surface. Further, the likelihood of delamination is exacerbated at high voltages and/or high temperatures, which may be at or greater than operating limitations of the MIM capacitor. The high voltages may, for example, be voltages greater than or equal to about 3.5 volts or some other suitable voltage, and/or the high temperatures may, for example, be temperatures greater than or equal to about 125 degrees Celsius or some other suitable temperature. Because the native oxide layer is prone to delamination, the MIM capacitor is prone to dielectric breakdown and has a low time-dependent dielectric breakdown (TDDB).

Aside from dielectric breakdown, the native oxide layer poses a number of other challenges. The native oxide layer has a high concentration of charge traps and a low energy band gap, such that the native oxide layer serves as a "step" for charge carriers to reach a sufficiently high energy level to pass through the insulator layer. This, in turn, aids charge carriers in passing through the insulator layer, increases leakage current, and increases capacitance-voltage (CV) dispersion. Further, the native oxide layer is dielectric and hence results in a parasitic capacitance in series with a primary capacitance of the insulator layer. Because of the series relationship, overall capacitance is degraded. Further yet, the native oxide layer allows an oxidant used to form the insulator layer to pass to the top surface of the bottom electrode and to further oxidize the bottom electrode. This enlarges the native oxide layer and makes it difficult to precisely control a thickness of the bottom electrode layer.

Various embodiments of the present application are directed towards a MIM capacitor comprising an enhanced interfacial layer to reduce breakdown failure, as well as a method for forming the MIM capacitor. In some embodiments, a bottom electrode layer is deposited over a substrate. A native oxide layer is formed on a top surface of the bottom electrode layer and has a first adhesion strength with the top surface of the bottom electrode layer. A plasma treatment process is performed to convert the native oxide layer and a top portion of the bottom electrode layer into an interfacial layer. The interfacial layer is conductive and has a second adhesion strength with the top surface of the bottom electrode layer that is greater than the first adhesion strength. An insulator layer is deposited on the interfacial layer, and a top electrode layer is deposited on the insulator layer. The bottom and top electrode layers, the interfacial layer, and the insulator layer are patterned to form a MIM capacitor. In some embodiments, the plasma treatment process is or comprises a nitrogen oxide (e.g., N_2O) plasma treatment process, the bottom electrode layer is or comprises titanium nitride, the interfacial layer is or comprises titanium oxynitride, or any combination of the foregoing.

By performing the plasma treatment, the top surface of the bottom electrode layer is smoothed. For example, ion bombardment from the plasma treatment may smooth the top surface of the bottom electrode layer. Further, the interfacial layer is formed in place of the native oxide layer and has a greater adhesion strength with the top surface of the bottom electrode layer than the native oxide layer. The greater adhesion strength, in turn, prevents delamination of the interfacial layer from the top surface of the bottom electrode layer. Hence, dielectric breakdown is reduced and TDDB is enhanced. Additionally, by performing the plasma treatment, negative effects of the native oxide layer may be reduced or eliminated. Charge traps in the native oxide layer may be repaired while the native oxide is converted to the interfacial layer. This, in turn, reduces leakage current and CV dispersion. Further, because the interfacial layer is conductive, the interfacial layer doesn't result in parasitic capacitance in series with a primary capacitance of the insulator layer. Hence, the interfacial layer doesn't degrade an overall capacitance of the MIM capacitor. Further yet, the interfacial layer may block diffusion of an oxidant used during formation of the insulator layer. This prevents the top surface of the bottom electrode layer from being oxidized and reducing a thickness of the bottom electrode layer. Further, if the top surface of the bottom electrode layer oxidized, the resulting oxide layer may lead to the same challenges described above for the native oxide layer.

With reference to FIG. 1, a cross-sectional view **100** of some embodiments of a metal-insulator-metal (MIM) capacitor **102** comprising a capacitor interfacial layer **104** is provided in which the capacitor interfacial layer **104** is enhanced to, among other things, reduce breakdown failure. The capacitor interfacial layer **104** overlies a bottom electrode **106**, a capacitor insulator layer **108** overlies the capacitor interfacial layer **104**, and a top electrode **110** overlies the capacitor insulator layer **108**. The top electrode **110** defines or is otherwise electrically coupled to a first terminal T1 of the MIM capacitor **102**, and the bottom electrode **106** defines or is otherwise electrically coupled to a second terminal T2 of the MIM capacitor.

The bottom electrode **106** and the top electrode **110** are conductive and may, for example, be or comprise titanium, titanium nitride, tantalum, tantalum nitride, some other suitable conductive material(s), or any combination of the foregoing. In some embodiments, a thickness Th_{be} of the bottom electrode **106** is about 150-400 angstroms, about 150-275 angstroms, or about 275-400 angstroms and/or a thickness Th_{te} of the top electrode **110** is about 400-600 angstroms, about 400-500 angstroms, or about 500-600 angstroms. Other thicknesses are, however, amenable for the bottom and top electrodes **106**, **110**. In some embodiments, a top surface of the bottom electrode **106** has a low average surface roughness. In some embodiments, surface roughness is quantified according to atomic force microscopy. Other processes are, however, amenable for quantifying surface roughness. In some embodiment, the low average surface roughness of the bottom electrode **106** is about 0.30-0.40 nanometers, about 0.30-0.35 nanometers, about 0.35-0.40 nanometers, about 0.36 nanometers, or about 0.34 nanometers according to atomic force microscopy. Other surface roughness values are, however, amenable.

The capacitor insulator layer **108** is dielectric and may, for example, be or comprise, zirconium oxide, aluminum oxide, hafnium oxide, silicon oxide, some other suitable dielectric(s), or any combination of the foregoing. In some embodiments, the capacitor insulator layer **108** is or comprises a metal oxide and/or is or comprises a high k

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dielectric. A high k dielectric may, for example, be a dielectric material having a dielectric constant greater than about 10 or some other suitable value.

The capacitor interfacial layer **104** is conductive throughout and has a high adhesion strength with the bottom electrode **106**. In some embodiments, adhesion strength is quantified by nanoscratch testing. Other processes are, however, amenable for quantifying adhesion. Nanoscratch testing may, for example, comprise applying a linearly increasing force to a layer (e.g., the capacitor interfacial layer **104**) on the bottom electrode **106** using a probe until adhesive failure. The force at which adhesive failure occurs represents adhesion strength. The more adhesive the layer is, the greater the force at which adhesive failure occurs. In some embodiment, the high adhesion strength of the capacitor interfacial layer **104** is about 1000-3000 micronewtons or greater than about 1000 micronewtons according to nanoscratch testing. Other adhesion values are, however, amenable. Below an adhesion of about 1000 micronewtons or some other suitable adhesion value, failure rates may be high. In some embodiments, the high adhesion strength of the capacitor interfacial layer **104** is greater than a native oxide layer would have with the bottom electrode **106**.

Because of the high adhesion strength, the likelihood of the capacitor interfacial layer **104** delaminating (i.e., reaching adhesive failure) is low. Hence, dielectric breakdown is reduced for the MIM capacitor **102** and TDDB is increased for the MIM capacitor **102**. Further, because the capacitor interfacial layer **104** is conductive, the capacitor interfacial layer **104** doesn't result in parasitic capacitance in series with a primary capacitance of the capacitor insulator layer **108**. Hence, the capacitor interfacial layer **104** doesn't degrade an overall capacitance of the MIM capacitor **102**. Note that two capacitors in series have a combined capacitance less than individual capacitances of the two capacitors.

The capacitor interfacial layer **104** may, for example, be or comprise titanium oxynitride, tantalum oxynitride, some other suitable conductive material(s), or any combination of the foregoing. In some embodiments, the capacitor interfacial layer **104** is or comprises TiO_xN_y . In some embodiments, x is greater than 0 and less than about 2 and y is less than about x. In some of such embodiments, y is also greater than 0. In alternative embodiments, x and y have other suitable values. In some embodiments, x decreases (i.e., oxygen decreases) from a top surface of the capacitor interfacial layer **104** to a bottom surface of the capacitor interfacial layer **104** and/or y increases (i.e., nitrogen increases) from the top surface to the bottom surface. The decrease and/or the increase may, for example, be continuous. In some embodiments, the bottom electrode **106** comprises or consists of a metal element and non-metal element and the capacitor interfacial layer **104** comprises or consists of oxygen, the metal element, and the non-metal element. The metal element may, for example, be titanium, tantalum, or some other suitable metal element, and/or the non-metal element may, for example, be nitrogen or some other suitable non-metal element. In alternative embodiments, the bottom electrode **106** consists of or consists essentially of the metal element and the capacitor interfacial layer **104** comprises or consists of oxygen, the metal element, and the non-metal element.

In some embodiments, the capacitor interfacial layer **104** is a diffusion barrier for oxidants. Oxidants may, for example, be used during formation of the capacitor insulator layer **108**. Absent the capacitor interfacial layer **104** being a diffusion barrier, oxidants could diffuse to the bottom electrode **106** and oxidize the top surface of the bottom electrode

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106 during formation of the capacitor insulator layer **108**. This would, in turn, reduce a thickness T_{be} of the bottom electrode **106** and would make it difficult to control the thickness T_{be} of the bottom electrode **106**. Further, a resulting oxide layer would create a parasitic capacitance in series with the primary capacitance of the capacitor insulator layer **108**. Because of the series relationship, overall capacitance of the MIM capacitor **102** would be degraded.

In some embodiments, the capacitor interfacial layer **104** has a thickness T_{int} between about 20-50 angstroms, about 20-35 angstroms, about 35-50 angstroms, or some other suitable thickness. If the thickness T_{int} of the capacitor interfacial layer **104** is too small (e.g., less than about 20 angstroms or some other suitable value), the capacitor interfacial layer **104** may be unable to block diffusion of oxidants during formation of the capacitor insulator layer **108**. Further, process limitations may limit the thickness T_{int} of the capacitor interfacial layer **104** to less than about 50 angstroms or some other suitable value. For example, attempting to form the capacitor interfacial layer **104** with a thickness more than about 50 angstroms using nitrogen oxide plasma treatment may result in a dielectric oxide layer. The dielectric oxide layer would induce parasitic capacitance in series with the primary capacitance of the capacitor insulator layer **108** and would degrade overall capacitance of the MIM capacitor **102**.

In some embodiments, the MIM capacitor **102** has low CV dispersion. CV dispersion may, for example, be quantified as

$$\frac{C - C_0}{C_0},$$

where C is capacitance of the MIM capacitor **102** at an operating voltage and C_0 is capacitance of the MIM capacitor **102** at 0 volts. Other approaches for quantifying CV dispersion are, however, amenable. The lower CV dispersion is, the more uniform capacitance of the MIM capacitor **102** is over a range of operating voltages. In some embodiment, the low CV dispersion of the MIM capacitor **102** is about 0.36, about 0.30, about 0.30-0.40, or less than about 0.35. Other CV dispersion values are, however, amenable. In some embodiments, the low CV dispersion of the MIM capacitor **102** is low compared to that of a MIM capacitor with a native oxide layer in place of the capacitor interfacial layer **104**.

The MIM capacitor **102** may, for example, have the low CV dispersion because the capacitor interfacial layer **104** is conductive. As a result, the capacitor interfacial layer **104** doesn't result in parasitic capacitance and doesn't facilitate leakage paths through the capacitor insulator layer **104**. This is in contrast with a MIM capacitor having a native oxide layer in place of the capacitor interfacial layer **104**. The native oxide layer results in a parasitic capacitance and further results in leakage paths that individually and/or collectively degrade CV dispersion. The parasitic capacitance results because the native oxide layer is dielectric. Leakage paths results because the native oxide layer has a high concentration of crystalline defects and a low energy band gap that provides a "step" for carriers to reach a sufficiently high energy level to pass through the capacitor insulator layer **108**.

With reference to FIG. 2A, a cross-sectional view **200A** of some embodiments of an integrated circuit (IC) chip comprising an interconnect structure **202** is provided in which

some first trench embodiments of the MIM capacitor **102** of FIG. **1** are embedded. The MIM capacitor **102** overlies a lower capacitor wire **204l** and has a downward protrusion defining a bottom electrode via (BEVA) **206**. An upper capacitor wire **204u** overlies the MIM capacitor **102**, and a top electrode via (TEVA) **208tv** extends from the upper capacitor wire **204u** to the MIM capacitor **102**. The lower capacitor wire **204l**, the upper capacitor wire **204u**, and the TEVA **208tv** are conductive and may be or comprise, for example, aluminum copper, aluminum, copper, some other suitable metal(s), or any combination of the foregoing.

The MIM capacitor **102**, the lower capacitor wire **204l**, the upper capacitor wire **204u**, and the TEVA **208tv** are surrounded by a plurality of intermetal dielectric (IMD) layers **210**. The IMD layers **210** are stacked upon each other and, in some embodiments, a plurality of etch stop layers **212** separate the IMD layers **210** from each other. In alternative embodiments, the etch stop layers **212** are omitted. The IMD layers **210** are a different material than the etch stop layers **212** and may, for example, be or comprise a low k dielectric and/or some other suitable dielectric(s). The etch stop layers **212** may, for example, be silicon carbide, some other suitable dielectric(s), or any combination of the foregoing.

With reference to FIG. **2B**, a cross-sectional view **200B** of some alternative embodiments of the IC chip of FIG. **2A** is provided in which the MIM capacitor **102** has a more symmetrical profile about a vertical axis at a width-wise center of the MIM capacitor **102**. Further, hard masks cover the MIM capacitor **102**. A top electrode hard mask **214** covers and has the same or substantially the same top layout as the top electrode **110**. A bottom electrode hard mask **216** covers and has the same or substantially the same top layouts as the bottom electrode **106**, the capacitor insulator layer **108**, and the capacitor interfacial layer **104**. Further, the bottom electrode hard mask **216** covers the top electrode hard mask **214**. In alternative embodiments, the capacitor insulator layer **108** instead has the same or substantially the same top layout as the top electrode hard mask **214**. The top and bottom electrode hard masks **214**, **216** may be or comprise, for example, silicon nitride and/or some other suitable dielectric(s).

In some embodiments, hard mask liners **218** are individual to the top and bottom electrode hard masks **214**, **216** and separate the top and bottom electrode hard masks **214**, **216** from the capacitor insulator layer **108** and the top electrode **110**. The hard mask liners **218** are different materials than the top and bottom electrode hard masks **214**, **216** and may be or comprise, for example, silicon oxide and/or some other suitable dielectric(s). In alternative embodiments, the hard mask liners **218** are omitted.

With reference to FIG. **2C**, a cross-sectional view **200C** of some alternative embodiments of the IC chip of FIG. **2B** is provided in which the top electrode **110** is indented at the BEVA **206**. Further, the bottom electrode **106**, the capacitor interfacial layer **104**, the capacitor insulator layer **108**, and the top electrode **110** have more curved edges. Further yet, the top electrode hard mask **214** and its corresponding hard mask liners **218** are omitted. In alternative embodiments, the top electrode hard mask **214** and its corresponding hard mask liners **218** remain on the top electrode **110** and separate the top electrode **110** from the bottom electrode hard mask **216** and its corresponding hard mask liners **218**.

With reference to FIG. **2D**, a cross-sectional view **200D** of some alternative embodiments of the IC chip of FIG. **2C** is provided in which the MIM capacitor **102** overlies a plurality of additional wires **204**. Further, an etch stop layer **212**

at the upper capacitor wire **204u** is omitted, and the TEVA **208tv** and the upper capacitor wire **204u** are integrated into a conductive structure **222**. Further yet, the conductive structure **222**, the lower capacitor wire **204l**, and the plurality of additional wires **204** are lined by interconnect barrier layers **224**. The interconnect barrier layers **224** are configured to prevent diffusion of material from the conductive structure **222**, the lower capacitor wire **204l**, and the plurality of additional wires **204** to underlying structure. In some embodiments, the conductive structure **222**, the lower capacitor wire **204l**, and the plurality of additional wires **204** are or comprise copper. In some embodiments, the interconnect barrier layers **224** are or comprise titanium, tantalum, titanium nitride, tantalum nitride, or some other suitable barrier material. Other materials are, however, amenable. In some embodiments, the interconnect barrier layers **224** and the capacitor interfacial layer **104** are or comprise a same material.

With reference to FIG. **2E**, a cross-sectional view **200E** of some alternative embodiments of the IC chip of FIG. **2A** are provided in which the BEVA **206** is omitted. Further, the capacitor insulator layer **108** cups an underside of the top electrode **110**, the capacitor interfacial layer **104** cups an underside of the capacitor insulator layer **108**, and the bottom electrode **106** cups an underside of the capacitor interfacial layer **104**. In some embodiments, the bottom electrode **106**, the capacitor interfacial layer **104**, the capacitor insulator layer **108**, and the top electrode **110** have U or V shaped profiles. Other profiles are, however, amenable.

With reference to FIGS. **3A** and **3B**, orthogonal cross-sectional views **300A**, **300B** of some alternative embodiments of the IC chip of FIG. **2A** is provided in which the upper capacitor wire **204u** is in a passivation layer **302** and an etch stop layer **212** at the upper capacitor wire **204u** is omitted. In alternative embodiments, the MIM capacitor **102** is replaced with the MIM capacitor **102** in any one of FIGS. **2B-2E** or some other suitable MIM capacitor. The cross-sectional view **300A** of FIG. **3A** is taken in an X direction and may, for example, be taken along line B-B' in FIG. **3B**. The cross-sectional view **300B** of FIG. **3B** is taken in a Y direction and may, for example, be taken along line A-A' in FIG. **3A**.

The passivation layer **302** overlies the IMD layers **210** and accommodates the upper capacitor wire **204u**. As seen in FIG. **3B**, the TEVA **208tv** extends from the upper capacitor wire **204u** to the MIM capacitor **102**. The passivation layer **302** is or comprises a different dielectric material than the IMD layers **210**. In some embodiments, the IMD layers **210** are or comprise a low k dielectric having a dielectric constant less than about 3.9, 3.0, or some suitable value, whereas the passivation layer **302** is or comprises a dielectric having a dielectric constant greater than the IMD layers **210**. In at least some of these embodiments, the passivation layer **302** has a dielectric constant between about 3.9-10 or greater than about 3.9. Other dielectric constants are, however, amenable for the IMD layers **210** and/or the passivation layer **302**.

With reference to FIG. **4A**, a cross-sectional view **400A** of some embodiments of an IC chip comprising a one-transistor one-capacitor (1T1C) cell **402** is provided in which the MIM capacitor **102** of FIG. **2A** is embedded. The MIM capacitor **102** overlies a substrate **404** in an interconnect structure **202**. The substrate **404** may be, for example, a bulk silicon substrate and/or some other suitable semiconductor substrate.

The interconnect structure **202** comprises an interlayer dielectric (ILD) layer **405**, an IMD layer **210**, and a passi-

vation layer 302 stacked over the substrate 404. The IMD layer 210 overlies the ILD layer 405, and the passivation layer overlies the IMD layer 210. The ILD layer 405 is a different material than the IMD layer 210 and may, for example, be or comprise silicon oxide and/or some other suitable dielectric. Further, the interconnect structure 202 comprises a plurality of wires 204 and a plurality of vias 208 stacked in the ILD, IMD, and passivation layers 405, 210, 302. The plurality of wires 204 and the plurality of vias 208 are conductive and define conductive paths leading from the MIM capacitor 102 and an underlying access transistor 406. A first conductive path leads from the MIM capacitor 102 to a bit line 204_{bl} above the MIM capacitor 102. A second conductive path leads from the MIM capacitor 102 to a drain region 408_d of the access transistor 406. A third conductive path leads from a source region 408_s of the access transistor 406 to a source line 204_{s1} above the source region 408_s. A fourth conductive path leads from a gate electrode 410 of the access transistor 406 to a word line 204_{wl} above the gate electrode 410. Note that while the word line 204_{wl} is shown with two separate segments on opposite sides of the drain region 408_d, the word line 204_{wl} may be continuous outside the cross-sectional view 300A.

The access transistor 406 comprises the drain and source regions 408_d, 408_s, and further comprises the gate electrode 410 and a gate dielectric layer 412. The drain and source regions 408_d, 408_s are in the substrate 404 and correspond to doped regions of the substrate 404. The gate electrode 410 overlies the gate dielectric layer 412 and is sandwiched between the drain and source regions 408_d, 408_s. In some embodiments, a sidewall spacer structure 414 is on sidewalls of the gate electrode 410 and/or the access transistor 406 is surrounded by a trench isolation structure 416. The sidewall spacer structure 414 and the trench isolation structure 416 are or comprise dielectric material(s). The access transistor 406 may, for example, be an insulated gate field-effect transistor (IGFET) or some other suitable transistor.

With reference to FIG. 4B, a cross-sectional view 400B of some alternative embodiments of the IC chip of FIG. 4A is provided in which the MIM capacitor 102 of FIG. 2A is replaced with the MIM capacitor 102 of FIG. 2B. In some other alternative embodiments of the IC chip of FIG. 4A, the MIM capacitor 102 of FIG. 2A is replaced with the MIM capacitor 102 in any one of FIGS. 2C-2E, 3A, and 3B or some other suitable MIM capacitor.

With reference to FIG. 5, a cross-sectional view 500 of some embodiments of an IC chip comprising a decoupling-capacitor region 502_{dc} and a logic region 502_l is provided in which multiple MIM capacitors 504 are in the decoupling-capacitor and logic regions 502_{dc}, 502_l and are each configured as the MIM capacitor 102 of FIG. 2B. In some alternative embodiments of the IC chip of FIG. 5, the MIM capacitors 504 are each configured as the MIM capacitor 102 in any one of FIGS. 2A, 2C-2E, 3A, and 3B or some other suitable MIM capacitor.

An interconnect structure 202 overlies a substrate 404. The interconnect structure 202 and/or the substrate 404 may, for example, be as described with regard to FIG. 4A. The interconnect structure 202 comprises an ILD layer 405, an IMD layer 210 overlying the ILD layer 405, and a passivation layer 302 overlying the IMD layer 210. Further, the interconnect structure 202 comprises a plurality of wires 204 and a plurality of vias 208 stacked in the ILD, IMD, and passivation layers 405, 210, 302. The plurality of wires 204 and the plurality of vias 208 are conductive and define conductive paths leading from the MIM capacitors 504 and also from multiple transistors 506 under the MIM capacitors

504. In some embodiments, there are no wires and vias, except for the illustrated wire 204, directly under the MIM capacitor 504 at the decoupling-capacitor region 502_{dc}.

The transistors 506 may, for example, each be configured as the access transistor 406 of FIG. 4A and/or may, for example, each be an IGFET or some other suitable transistor. The transistors 506 comprise individual source/drain regions 508_i, individual gate electrodes 410, and individual gate dielectric layers 412. Further, two of the transistors 506 that neighbor each other share a shared source/drain region 508_s. The gate electrodes 410 overlie the gate dielectric layers 412 and are each sandwiched between two of the individual and/or shared source/drain regions 508_i, 508_s. In some embodiments, sidewall spacer structures 414 are individual to the gate electrodes 410 and line sidewalls of the gate electrodes 410. In some embodiments, the transistors 506 are surrounded and separated by a trench isolation structure 416. In some embodiments, there are no transistors and/or other semiconductor devices on the substrate 404 and directly under the MIM capacitor 504 at the decoupling-capacitor region 502_{dc}.

While the MIM capacitor 102 in the IC chip of FIG. 4A or 4B is shown as being between the fourth and fifth wire levels, the MIM capacitor 102 may be between any other neighboring wire levels in alternative embodiments. Similarly, while the MIM capacitors 504 in the IC chip of FIG. 5 are shown as being between the fourth and fifth wire levels, the MIM capacitors 504 may be between any other neighboring wire levels in alternative embodiments. Further, while the MIM capacitor 102 in the IC chip of FIG. 4A or 4B is shown as being in the IMD layer 210, the MIM capacitor 102 may be in the passivation layer 302 or the ILD layer 405 in alternative embodiments. Similarly, while the MIM capacitors 504 in the IC chip of FIG. 5 are shown as being in the IMD layer 210, the MIM capacitors 504 may be in the passivation layer 302 or the ILD layer 405 in alternative embodiments.

With reference to FIG. 6A, a cross-sectional view 600A of some embodiments of an IC chip comprising an interconnect structure 202 is provided in which planar embodiments of the MIM capacitor 102 of FIG. 1 are embedded. The planar embodiments of the MIM capacitor 102 are to be contrasted with the first trench embodiments of the MIM capacitor 102 in FIGS. 2A-2E. The MIM capacitor 102 fills a trench (e.g., to define the BEVA 206) in the first trench embodiments, whereas the MIM capacitor 102 does not in FIG. 6A. As such, the MIM capacitor 102 of FIG. 6A has a planar or substantially planar bottom profile.

The MIM capacitor 102 underlies a first upper capacitor wire 204_{u1} and a second upper capacitor wires 204_{u2}. A first capacitor via 208_{fc} extends from the top electrode 110 to the first upper capacitor wire 204_{u1} to electrically couple the first upper capacitor wires 204_{u1} to the top electrode 110. Similarly, a second capacitor via 208_{sc} extends from the bottom electrode 106 to the second upper capacitor wire 204_{u2} to electrically couple the second upper capacitor wire 204_{u2} to the bottom electrode 106. A plurality of IMD layers 210 surround the MIM capacitor 102, the first upper capacitor wire 204_{u1}, and the second upper capacitor wire 204_{u2}. In some embodiments, etch stop layers 212 separate the IMD layers 210 from each other. In alternative embodiments, the etch stop layers 212 are omitted.

With reference to FIG. 6B, a cross-sectional view 600B of some alternative embodiments of the IC chip of FIG. 6A is provided in which the top electrode 110 has a stepped profile. Further, the MIM capacitor 102 is in a passivation layer 302 that overlies an IMD layer 210, and the first and

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second capacitor vias **208/c**, **208sc** extend through and electrically couple with the MIM capacitor **102**. The first capacitor via **208/c** extends through the top electrode **110**, from the first upper capacitor wire **204u₁** to a first lower capacitor wire **204l₁**. Similarly, the second capacitor via **208sc** extends through the bottom electrode **106**, from the second upper capacitor wire **204u₂** to a second lower capacitor wire **204l₂**.

With reference to FIG. 6C, a cross-sectional view **600C** of some alternative embodiments of the IC chip of FIG. 6B is provided in which the MIM capacitor **102** further comprises a middle electrode **602**. The middle electrode **602** is between the bottom and top electrodes **106**, **110**. A first capacitor interfacial layer **104a** is on a top surface of the bottom electrode **106**, and a second capacitor interfacial layer **104b** is on a top surface of the middle electrode **602**. A first capacitor insulator layer **108a** is between the bottom and middle electrodes **106**, **602** and separates the first capacitor interfacial layer **104a** and the bottom electrode **106** from middle electrode **602** and the top electrode **110**. A second capacitor insulator layer **108b** is between the top electrode **110** and the bottom electrode **106** and also between the top electrode **110** and the middle electrode **602**. Further, the second capacitor insulator layer **108b** separates the second capacitor interfacial layer **104b**, the middle electrode **602**, and the first capacitor insulator layer **108a** from the top electrode **110**.

The bottom electrode **106** and the first capacitor interfacial layer **104a** are respectively as the bottom electrode **106** and the capacitor interfacial layer **104** are described with regard to FIG. 1. Similarly, the middle electrode **602** and the second capacitor interfacial layer **104b** are respectively as the bottom electrode **106** and the capacitor interfacial layer **104** are described with regard to FIG. 1. As such, the first and second capacitor interfacial layers **104a**, **104b** reduce dielectric breakdown and enhance TDDB of the MIM capacitor **102**. The first and second capacitor insulator layers **108a**, **108b** and the top electrode **110** are respectively as the capacitor insulator layer **108** and the top electrode **110** are described with regard to FIG. 1.

The first capacitor via **208/c** extends through the middle electrode **602**, from the first upper capacitor wire **204u₁** to the first lower capacitor wire **204l₁**, and electrically couples with the middle electrode **602**. Similarly, the second capacitor via **208sc** extends through the bottom and top electrodes **106**, **110**, from the second upper capacitor wire **204u₂** to the second lower capacitor wire **204l₂**, and electrically couples with the bottom and top electrodes **106**, **110**. Because the second capacitor via **208sc** electrically couples to both the bottom and top electrodes **106**, **110** and the bottom and top electrodes **106**, **110** are respectively under and over the middle electrode **602**, the MIM capacitor can be modeled as two capacitors in parallel. A first capacitor is defined between the bottom and middle electrodes **106**, **602**, and a second capacitor is defined between the top and middle electrodes **110**, **602**.

With reference to FIG. 7, a cross-sectional view **700** of some alternative embodiments of a portion of the IC chip of FIG. 6C at the first capacitor via **208/c** is provided in which the MIM capacitor **102** overlies a plurality of additional wires **204**. Further, the first capacitor via **208/c** and the first upper capacitor wire **204u₁** are continuous with each other and integrated into a conductive structure **702** having rounded corners and uneven surfaces. Further yet, the passivation layer **302** covers the conductive structure **702** and has a top surface that substantially conforms to the conductive structure **702**. Further yet, the first lower capacitor wire

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204l₁, the conductive structure **702**, and the plurality of additional wires **204** are lined by interconnect barrier layers **224**. The interconnect barrier layers **224** are configured to prevent diffusion of material from the first lower capacitor wire **204l₁**, the conductive structure **702**, and the plurality of additional wires **204** to underlying structure.

With reference to FIGS. 8A and 8B, orthogonal cross-sectional views **800A**, **800B** of some alternative embodiments of a portion of the IC chip of FIG. 7 is provided in which the conductive structure **702** has a top indent **704** extending to the first capacitor via **208/c**. The cross-sectional view **800A** of FIG. 8A is taken in an X direction and may, for example, be taken along line D-D' in FIG. 8B. The cross-sectional view **800B** of FIG. 8B is taken in a Y direction and may, for example, be taken along line C-C' in FIG. 8A.

With reference to FIG. 9, a cross-sectional view **900** of some alternative embodiments of the IC chip of FIG. 8B is provided in which additional wires **204** are below the first upper capacitor wire **204u₁** and the MIM capacitor **102**.

With reference to FIG. 10, a cross-sectional view **1000** of some embodiments of an IC chip comprising a 1T1C cell **402** is provided in which the MIM capacitor **102** of FIG. 6C is embedded. The 1T1C **402** may, for example, be as in FIG. 4A, except for inclusion of the MIM capacitor **102** of FIG. 6C instead of the MIM capacitor of FIG. 2A. In some alternative embodiments of the IC chip, the MIM capacitor **102** of FIG. 6C is replaced with the MIM capacitor **102** in any one of FIGS. 6A and 6B or some other suitable MIM capacitor. In some alternative embodiments of the IC chip in which the MIM capacitor **102** of FIG. 6C is replaced with the MIM capacitor of FIG. 6A, the MIM capacitor **102** of FIG. 6A is located in the IMD layer **210** instead of the passivation layer **302** as shown in FIG. 6A.

With reference to FIG. 11, a cross-sectional view **1100** of some embodiments of an IC chip comprising a decoupling-capacitor region **502dc** and a logic region **502l** is provided in which multiple MIM capacitors **504** are in the decoupling-capacitor and logic regions **502dc**, **502l** and are each configured as the MIM capacitor **102** of FIG. 6C. The decoupling-capacitor region **502dc** and the logic region **502l** may, for example, be as in FIG. 5, except that the MIM capacitors **504** are each configured as in FIG. 6C instead of as in FIG. 2B. In some alternative embodiments of the IC chip, the MIM capacitors **504** are each configured as the MIM capacitor **102** in any one of FIGS. 6A and 6B or some other suitable MIM capacitor.

While the MIM capacitor **102** in the IC chip of FIG. 10 is shown as being between the fifth and sixth wire levels, the MIM capacitor **102** may be between any other neighboring wire levels in alternative embodiments. Similarly, while the MIM capacitors **504** in the IC chip of FIG. 11 are shown as being between the fifth and sixth wire levels, the MIM capacitors **504** may be between any other neighboring wire levels in alternative embodiments. Further, while the MIM capacitor **102** in the IC chip of FIG. 10 is shown as being in the passivation layer **302**, the MIM capacitor **102** may be in the IMD layer **210** or the ILD layer **405** in alternative embodiments. Similarly, while the MIM capacitors **504** in the IC chip of FIG. 11 are shown as being in the passivation layer **302**, the MIM capacitors **504** may be in the IMD layer **210** or the ILD layer **405** in alternative embodiments.

With reference to FIG. 12A, a cross-sectional view **1200A** of some embodiments of an IC chip comprising a substrate **404** is provided in which second trench embodiments of the MIM capacitor **102** of FIG. 1 are embedded. The second trench embodiments are to be contrasted with the first trench

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embodiments in FIGS. 2A-2E. In the first trench embodiments, the MIM capacitor 102 is in a trench in an interconnect structure 202 overlying the substrate 404. In the second trench embodiments, the MIM capacitor 102 is in a trench in the substrate 404.

The MIM capacitor 102 comprises a plurality of trench electrodes 1202 and a substrate electrode 1204. In alternative embodiments, the substrate electrode 1204 is omitted. The trench electrodes 1202 define one or more trench segments 1206 protruding into the substrate 404, whereas the substrate electrode 1204 corresponds to a doped region of the substrate 404 that surrounds the trench segment(s) 1206. The trench electrodes 1202 overlie the substrate electrode 1204 and are vertically stacked. Further, widths of the trench electrodes 1202 increase from a top of the vertical stack to a bottom of the vertical stack.

A plurality of capacitor interfacial layers 104 and a plurality of capacitor insulator layers 108 are also stacked with the trench electrodes 1202 and the substrate electrode 1204. The capacitor insulator layers 108 separate the trench electrodes 1202 from each other and from the substrate electrode 1204. The capacitor insulator layers 108 may, for example, be as described with regard to FIG. 1. Except for the topmost one of the trench electrodes 1202, the trench electrodes 1202 have top surfaces covered respectively by the capacitor interfacial layers 104. The capacitor interfacial layers 104 may, for example, be as described with regard to FIG. 1 and, except for the topmost one of the trench electrodes 1202, the trench electrodes 1202 may, for example, be as the bottom electrode 106 of FIG. 1 is described. Hence, the capacitor interfacial layers 104 have high adhesion to the trench electrodes 1202 and reduce dielectric breakdown. The topmost one of the trench electrodes 1202 may, for example, be as the top electrode 110 of FIG. 1 is described. In some embodiments, the capacitor insulator layers 108 and/or the capacitor interfacial layers 104 each has a same top layout as an immediately underlying electrode (e.g., a substrate or trench electrode). In some embodiments, a cap layer 1208 overlies the topmost one of the trench electrodes 1202 and fills gaps at the trench segments 1206.

The interconnect structure 202 overlies the MIM capacitor 102 and comprises a first upper capacitor wire 204_{u1} and a second upper capacitor wire 204_{u2}. Further, the interconnect structure 202 comprises a plurality of capacitor vias 208_c. The capacitor vias 208_c electrically couple every other electrode of the MIM capacitor 102, from a bottom of the MIM capacitor 102 to a top of the MIM capacitor 102, to the second upper capacitor wire 204_{u2}. Further, the capacitor vias 208_c electrically couple remaining electrodes of the MIM capacitor 102 to the first upper capacitor wire 204_{u1}. An ILD layer 405 surrounds the MIM capacitor 102 and the capacitor vias 208_c, and an IMD layer 210 overlies the ILD layer 405 and surrounds the first and second upper capacitor wires 204_{u1}, 204_{u2}.

With reference to FIG. 12B, a cross-sectional view 1200B of some alternative embodiments of the IC chip of FIG. 12A is provided in which edges of the MIM capacitor 102 have a more rounded profile.

With reference to FIG. 13, an expanded cross-sectional view 1300 of some embodiments of the IC chip of FIG. 12A is provided in which more of the interconnect structure 202 is shown and multiple transistors 506 neighbor the MIM capacitor 102 on the substrate 404. In some alternative embodiments of the IC chip, the MIM capacitor 102 of FIG. 12A is replaced with the MIM capacitor 102 of FIG. 12B or some other suitable MIM capacitor. The interconnect struc-

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ture 202 comprises the ILD layer 405, the IMD layer 210 overlying the ILD layer 405, and a passivation layer 302 overlying the IMD layer 210. Further, the interconnect structure 202 comprises a plurality of wires 204 and a plurality of vias 208 stacked in the ILD, IMD, and passivation layers 405, 210, 302. The wires 204 and the vias 208 are conductive and define conductive paths leading from the MIM capacitors 504 and also from the transistors 506.

The transistors 506 may, for example, each be configured as the transistor 506 of FIG. 5 and/or may, for example, each be an IGFET or some other suitable transistor. The transistors 506 comprise individual source/drain regions 508_i, individual gate electrodes 410, and individual gate dielectric layers 412. The gate electrodes 410 overlie the gate dielectric layers 412 and are each sandwiched between two of the individual source/drain regions 508_i. In some embodiments, sidewall spacer structures 414 are individual to the gate electrodes 410 and line sidewalls of the gate electrodes 410. In some embodiments, the transistors 506 and/or the MIM capacitor 102 are surrounded and separated by a trench isolation structure 416.

With reference to FIGS. 14-22, a series of cross-sectional views 1400-2200 of some embodiments of a method for forming an IC chip comprising a MIM capacitor is provided in which the MIM capacitor comprises an interfacial layer enhanced to reduce breakdown failure. The IC chip formed by the method may, for example, be the IC chip of FIG. 4A and/or the MIM capacitor may, for example, be as described with regard to FIG. 2A.

As illustrated by the cross-sectional view 1400 of FIG. 14, a trench isolation structure 416 and an access transistor 406 are formed on the substrate 404. The trench isolation structure 416 is formed extending into a top of the substrate 404 and has a pair of segments that are laterally spaced. The trench isolation structure 416 may, for example, be a shallow trench isolation (STI) structure or some other suitable trench isolation structure. The access transistor 406 is formed between the segments of the trench isolation structure 416 after the trench isolation structure 416 is formed. The access transistor 406 comprises a gate dielectric layer 412, a gate electrode 410 overlying the gate dielectric layer 412, and a sidewall spacer structure 414 on sidewalls of the gate electrode 410. Further, the access transistor 406 comprises a source region 408_s and a drain region 408_d between which the gate electrode 410 is sandwiched. The access transistor 406 may, for example, be an IGFET or some other suitable transistor.

Also illustrated by the cross-sectional view 1400 of FIG. 14, an interconnect structure 202 is partially formed over and electrically coupled to the access transistor 406. The interconnect structure 202 comprises an ILD layer 405 and a first IMD layer 210_a overlying the ILD layer 405. Further, the interconnect structure 202 comprises a plurality of wires 204 and a plurality of vias 208 stacked in the ILD layer 405 and the first IMD layer 210_a. The plurality of wires 204 and the plurality of vias 208 define a conductive path leading from the access transistor 406 to a lower capacitor wire 204₁ of the plurality of wires 204. The lower capacitor wire 204₁ is at a top of the interconnect structure 202 and provides a base for a MIM capacitor hereafter formed on the lower capacitor wire 204₁.

As illustrated by the cross-sectional view 1500 of FIG. 15, a second IMD layer 210_b is deposited on the lower capacitor wire 204₁. For drawing compactness, a portion of the interconnect structure 202 underlying the lower capacitor wire 204₁ is not shown herein (i.e., in FIG. 15) and hereafter (i.e., in FIGS. 16-22). In some alternative embodiments, an

etch stop layer (e.g., **212** in FIG. 2A) is first deposited on the lower capacitor wire **2041** to separate the second IMD layer **210b** from the lower capacitor wire **2041**.

As illustrated by the cross-sectional view **1600** of FIG. 16, the second IMD layer **210b** is patterned to form a trench **1602** overlying and exposing the lower capacitor wire **2041**. The patterning may, for example, be performed by a photolithography/etching process or some other suitable etching process.

As illustrated by the cross-sectional view **1700** of FIG. 17, a bottom electrode layer **1702** is deposited on the second IMD layer **210b** and lining the trench **1602** in a first process chamber **1704**. In some embodiments, the bottom electrode layer **1702** is deposited by thermal ALD. In alternative embodiments, the bottom electrode layer **1702** is deposited by PVD or some other suitable deposition process. The bottom electrode layer **1702** is conductive and may, for example, be or comprise titanium, titanium nitride, tantalum, tantalum nitride, some other suitable conductive material(s), or any combination of the foregoing. In some embodiments, a thickness Th_{be} of the bottom electrode layer **1702** is about 150-400 angstroms, about 150-275 angstroms, or about 275-400 angstroms. Other thicknesses are, however, amenable for the bottom electrode layer **1702**.

As illustrated by the cross-sectional view **1800** of FIG. 18, the bottom electrode layer **1702** and structure underlying the bottom electrode layer **1702** are transferred to a second process chamber **1802**. In doing so, the top surface **1702t** of the bottom electrode layer **1702** is exposed to air and oxidizes. The oxidation partially consumes the bottom electrode layer **1702** to recess the top surface **1702t** of the bottom electrode layer **1702** and decrease the thickness Th_{be} of the bottom electrode layer **1702**. Further, the oxidation increases an average roughness of the top surface **1702t** of the bottom electrode layer **1702** and forms a native oxide layer **1804** on the top surface **1702t**. The increased average surface roughness degrades electrical field uniformity for a MIM capacitor hereafter formed from the bottom electrode layer **1702** and may lead to non-uniformity in operating parameters of the MIM capacitor when manufactured in bulk.

In some embodiments, surface roughness is quantified according to atomic force microscopy. Other processes are, however, amenable. In some embodiment, the average surface roughness of the bottom electrode layer **1702** is about 0.70-0.80 nanometers, about 0.72 nanometers, greater than about 0.70 nanometers, or some other suitable value according to atomic force microscopy. In some embodiments, because of the increase in average surface roughness, total thickness variation (TTV) of the bottom electrode layer **1702** is increased. TTV is a difference between a smallest value of the thickness Th_{be} of the bottom electrode layer **1702** and a largest value of the thickness Th_{be} .

The native oxide layer **1804** has low adhesion with the bottom electrode layer **1702**. In some embodiments, adhesion strength is quantified by nanoscratch testing. Other processes are, however, amenable for quantifying adhesion. Nanoscratch testing may, for example, comprise applying a linearly increasing force to a layer (e.g., the native oxide layer **1804**) using a probe until adhesive failure. The force at which adhesive failure occurs represents adhesion strength. The more adhesive the layer is, the greater the force at which adhesive failure occurs. In some embodiment, the low adhesion strength of the native oxide layer **1804** is less than about 500 or 1000 micronewtons according to nanoscratch testing. Other adhesion values are, however, amenable. Below an adhesion of about 1000 micronewtons or some

other suitable adhesion value, delamination and dielectric breakdown may be high, especially at high temperatures and/or high voltages.

As illustrated by the cross-sectional view **1900** of FIG. 19, the native oxide layer **1804** and the top surface **1702t** of the bottom electrode layer **1702** are plasma treated in the second process chamber **1802** to form an interfacial layer **1902** in place of the native oxide layer **1804**. The plasma treatment converts the native oxide layer **1804** and an upper portion of the bottom electrode layer **1702** into the interfacial layer **1902**. As such, the plasma treatment recesses the top surface **1702t** of the bottom electrode layer **1702** and decreases the thickness Th_{be} of the bottom electrode layer **1702**. Further, ion bombardment from the plasma treatment erodes the native oxide layer **1804** and smooths the top surface **1702t** of the bottom electrode layer **1702**, thereby improving electric field uniformity for a MIM capacitor hereafter formed.

The plasma treatment comprises generating plasma **1904** from a process gas and subsequently exposing the native oxide layer **1804** and the top surface **1702t** of the bottom electrode layer **1702** to the plasma **1904**. The process gas and/or ions from the plasma **1904** diffuse into the native oxide layer **1804** and/or the bottom electrode layer **1702** to form the interfacial layer **1902**. In some embodiments, the process gas comprises, consists essentially of, or consists of nitrous oxide. In alternative embodiments, the process gas comprises, consists essentially of, or consists of ammonia and/or ozone. In some embodiments, the plasma treatment generates the plasma **1904** using: 1) a power of about 200-1900 watts, about 200-550 watts, or some other suitable value; 2) a pressure of about 1.0-3.0 torr, about 1.0-2.0 torr, or some other suitable value; 3) a flow rate for the process gas that is about 10000-20000 standard cubic centimeters per minute (SCCM), about 10000-14000 SCCM, or some other suitable value; 4) or any combination of the foregoing. In some embodiments, the plasma treatment has one or more cycles and each of the cycles exposes the native oxide layer **1804** and the top surface **1702t** of the bottom electrode layer **1702** to the plasma **1904** for about 5-60 seconds, about 5-20 seconds, or some other suitable value. In some embodiments, the cycles have a 25%, 50%, or 75% duty cycle, where ON time corresponds to exposure and OFF time corresponds to no exposure.

In some embodiment, the process treatment comprises: 1) generating the plasma **1904** at a power of about 1800 watts, a pressure of about 2.5 torr, and a nitrous-oxide flow rate of about 18000 SCCM; and 2) exposing the native oxide layer **1804** and the bottom electrode layer **1702** to the plasma for about 45 seconds. In some embodiment, the process treatment comprises: 1) generating the plasma **1904** at a power of about 500 watts, a pressure of about 1.6 torr, and a nitrous-oxide flow rate of about 12000 sccm; and 2) exposing the native oxide layer **1804** and the bottom electrode layer **1702** to the plasma for about 10 seconds. In some embodiments, the process treatment has four cycles and comprises: 1) generating the plasma **1904** at a power of about 400-600 watts or about 500 watts; and 2) exposing the native oxide layer **1804** and the bottom electrode layer **1702** to the plasma **1904** for about 40-50 seconds, or about 45 seconds, for each cycle. In some embodiments, the process treatment has two cycles and comprises: 1) generating the plasma **1904** at a power of about 200-300 watts or about 250 watts; and 2) exposing the native oxide layer **1804** and the bottom electrode layer **1702** to the plasma **1904** for about 5-15 seconds, or about 10 seconds, for each cycle.

Because of the plasma treatment, which may be performed at relatively low cost, the interfacial layer **1902** is conductive throughout, from a topmost surface of the interfacial layer **1902** to a bottommost surface of the interfacial layer **1902**. Further, the interfacial layer **1902** has a high adhesion strength with the bottom electrode layer **1702** compared to the native oxide layer **1804**. In some embodiment, the high adhesion strength is about 1000-3000 micronewtons or greater than about 1000 micronewtons according to nanoscratch testing. Other adhesion values are, however, amenable. Below an adhesion of about 1000 micronewtons or some other suitable adhesion value, delamination of the interfacial layer **1902** may be high. Hence, because of the high adhesion strength, the likelihood of the interfacial layer **1902** delaminating (i.e., reaching adhesive failure) is low. This, in turn, reduces dielectric breakdown for a MIM capacitor being formed and increases TDDDB for the MIM capacitor.

The interfacial layer **1902** may, for example, be or comprise titanium oxynitride, tantalum oxynitride, some other suitable conductive material(s), or any combination of the foregoing. In some embodiments, the interfacial layer **1902** is or comprises TiO_xN_y . In some embodiments, x is greater than 0 and less than about 2 and y is less than about x . In some of such embodiments, y is also greater than 0. In alternative embodiments, x and y have other suitable values. In some embodiments, x decreases from a top surface of the interfacial layer **1902** to a bottom surface of the interfacial layer **1902** and/or y increases from the top surface to the bottom surface. The decrease and/or the increase may, for example, be continuous. In some embodiments, the bottom electrode layer **1702** comprises or consists of a metal element and non-metal element and the interfacial layer **1902** comprises or consists of oxygen, the metal element, and the non-metal element. The metal element may, for example, be titanium, tantalum, or some other suitable metal element, and/or the non-metal element may, for example, be nitrogen or some other suitable non-metal element. In alternative embodiments, the bottom electrode layer **1702** consists of or consists essentially of the metal element and the interfacial layer **1902** comprises or consists of oxygen, the metal element, and the non-metal element.

In some embodiments, the interfacial layer **1902** is a diffusion barrier for at least an oxidant used during a subsequent deposition of an insulator layer. In some embodiments, the interfacial layer **1902** has a thickness Th_{int} between about 20-50 angstroms, about 20-35 angstroms, about 35-50 angstroms, or some other suitable thickness. If the thickness Th_{int} is too small (e.g., less than about 20 angstroms or some other suitable value), the interfacial layer **1902** may be unable to block diffusion of the oxidant during the subsequent deposition of the insulator layer. Further, process limitations may limit the thickness Th_{int} to less than about 50 angstroms or some other suitable value. When the thickness Th_{int} of the interfacial layer **1902** becomes too large (e.g., greater than about 50 angstroms or some other suitable value), crystalline quality is poor and hence adhesion to the bottom electrode layer **1702** is poor. Further, a portion of the interfacial layer **1902** may become dielectric. For example, when the bottom electrode layer **1702** is titanium nitride, titanium oxide may form. The dielectric portion results in a parasitic capacitance in series with a main capacitance of the MIM capacitor being formed. Because of the series relationship, the overall capacitance of the MIM capacitor is degraded. Further, the dielectric portion may have a low energy band gap that increases leakage current in the MIM capacitor and increases CV dispersion.

As illustrated by the cross-sectional view **2000** of FIG. **20**, an insulator layer **2002** is deposited on the interfacial layer **1902**. The insulator layer **2002** is dielectric and may, for example, be or comprise, zirconium oxide, aluminum oxide, hafnium oxide, silicon oxide, some other suitable dielectric(s), or any combination of the foregoing. In some embodiments, the insulator layer **2002** is or comprises a metal oxide and/or is or comprises a high k dielectric. A high k dielectric may, for example, be a dielectric material having a dielectric constant greater than about 10 or some other suitable value. The insulator layer **2002** may, for example, be deposited by ALD or some other suitable deposition process.

In at least some embodiments in which the insulator layer **2002** is deposited by ALD, the interfacial layer **1902** is exposed to an oxidant during the deposition. The oxidant may, for example, be or comprise ozone, water vapor, or some other suitable oxidant. As noted above, in some embodiments, the interfacial layer **1902** is a diffusion barrier for the oxidant. Absent the interfacial layer **1902** being a diffusion barrier for the oxidant, the oxidant could diffuse to the bottom electrode layer **1702** and oxidize the top surface **1702t** of the bottom electrode layer **1702** during deposition of the insulator layer **2002**. The oxidation would reduce a thickness T_{be} of the bottom electrode layer **1702** and would hence make it difficult to control the thickness T_{be} . For example, the thickness T_{be} of the bottom electrode layer **1702** may be reduced by an uncontrollable amount. As another example, roughness of the top surface **1702t** of the bottom electrode layer **1702** may be increased and/or TTV of the bottom electrode layer **1702** may be increased. Additionally, the oxidation would form an oxide layer on the bottom electrode layer **1702**. For example, where the bottom electrode layer **1702** is or comprise titanium nitride, the oxidation may form a titanium oxide layer. The oxide layer would be dielectric and create a parasitic capacitance in series with the primary capacitance of a MIM capacitor being formed. Because of the series relationship, overall capacitance of the MIM capacitor would be degraded. Further, the oxide layer may have a low energy band gap that acts as a step for charge carriers to reach a sufficiently high energy level to pass through the insulator layer **2002**. Hence, the oxide layer may increase leakage current in a MIM capacitor being formed. Additionally, the increased leakage current may increase CV dispersion of the MIM capacitor.

Also illustrated by the cross-sectional view **2000** of FIG. **20**, a top electrode layer **2004** is deposited over the insulator layer **2002**, filling a remainder of the trench **1602** (see FIG. **19**). The top electrode layer **2004** may, for example, be deposited by thermal ALD, PVD, or some other suitable deposition process. The top electrode layer **2004** is conductive and may, for example, be or comprise titanium, titanium nitride, tantalum, tantalum nitride, some other suitable conductive material(s), or any combination of the foregoing. In some embodiments, the top electrode layer **2004** is or comprises a same material as the bottom electrode layer **1702**. In some embodiments, a thickness Th_e of the top electrode layer **2004** is 400-600 angstroms, about 400-500 angstroms, about 500-600 angstroms, or some other suitable thickness.

In some embodiments, an anneal is performed to the top electrode layer **2004** after depositing the top electrode layer **2004** and before the acts described hereafter at FIG. **21**. The anneal may, for example, be or comprise a rapid thermal anneal (RTA) or some other suitable anneal. The anneal may, for example, be performed at about 350-450 degrees Celsius, about 400 degrees Celsius, or some other suitable

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temperature. The anneal may, for example, be performed for about 8-12 hours, about 10 hours, or some other suitable duration. In some embodiments, the anneal is or comprise an RTA performed at about 400 degrees for about 10 hours.

As illustrated by the cross-sectional view **2100** of FIG. **21**, the top electrode layer **2004** (see, e.g., FIG. **20**), the insulator layer **2002** (see, e.g., FIG. **20**), the interfacial layer **1902** (see, e.g., FIG. **20**), and the bottom electrode layer **1702** (see, e.g., FIG. **20**) are patterned to form a MIM capacitor **102** overlying the lower capacitor wire **2041**. The MIM capacitor **102** comprises a bottom electrode **106**, a capacitor interfacial layer **104**, a capacitor insulator layer **108**, and a top electrode **110** vertically stacked. The patterning may, for example, be performed by a photolithography/etching process or some other suitable etching process.

As illustrated by the cross-sectional view **2200** of FIG. **22**, the interconnect structure **202** is completed around the MIM capacitor **102**. Upon completion, the interconnect structure **202** comprises a third IMD layer **210c** overlying the MIM capacitor **102** and further comprises a passivation layer **302** overlying the third IMD layer **210c**. Further, the interconnect structure **202** comprises a plurality of additional wires **204** and a plurality of additional vias **208** in the third IMD layer **210c** and the passivation layer **302**. The plurality of additional wires **204** comprises an upper capacitor wire **204u**, and the plurality of additional vias **208** comprises a TEVA **208/v** extending from the upper capacitor wire **204u** to the top electrode **110**.

While not shown, breakdown failure testing may be performed on the MIM capacitor **102** after completing the interconnect structure **202**. The breakdown failure testing applies a high voltage across the MIM capacitor **102**, from the bottom electrode **106** to the top electrode **110**. The high voltage may, for example, be a voltage greater than about 3.5 volts, between about 3.0-4.0 volts, or some other suitable voltage. In some embodiments, an ambient environment of the MIM capacitor **102** has a temperature of about 125 degrees Celsius, 100-150 degrees Celsius, greater than about 100 degrees Celsius, or some other suitable temperature. Because the capacitor interfacial layer **104** has high adhesion with the bottom electrode **106**, the likelihood of the capacitor interfacial layer **104** delaminating from the bottom electrode **106** is low even at the high electric fields induced by the high voltage. Accordingly, the likelihood of dielectric breakdown is low and TDDB is high for the MIM capacitor **102**. For example, TDDB may be greater than about 10 years or some other suitable duration. As another example, there may be less than about 0.001 failures per centimeter squared, or some other suitable number of failures per centimeter squared, when the MIM capacitor **102** is manufactured in bulk.

While FIGS. **14-22** are described with reference to a method, it will be appreciated that the structures shown in FIGS. **14-22** are not limited to the method but rather may stand alone separate of the method. While FIGS. **14-22** are described as a series of acts, it will be appreciated that the order of the acts may be altered in other embodiments. While FIGS. **14-22** illustrate and describe as a specific set of acts, some acts that are illustrated and/or described may be omitted in other embodiments. Further, acts that are not illustrated and/or described may be included in other embodiments. For example, while the illustrated and/or described acts form embodiments of the MIM capacitor **102** in FIG. **2A**, illustrated and/or describes acts may be omitted and/or non-illustrated acts may be included to form the MIM capacitor **102** according to embodiments in any one of FIGS. **2B-2E**, **6A-6C**, **12A**, and **12B**. In some embodiments, to

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form the MIM capacitor **102** according to embodiments in FIGS. **6A-6C**, the patterning at FIG. **16** is omitted. In some embodiments, to form the MIM capacitor **102** according to embodiments in FIGS. **21A** and **12B**, the acts at FIGS. **14** and **15** are omitted and the patterning at FIG. **16** is performed into the substrate **404**.

With reference to FIG. **23**, a block diagram **2300** of some embodiments of the method of FIGS. **14-22** is provided. The method may, for example, be employed to form a MIM capacitor according to the embodiments at FIGS. **2A-2E**. Further, alternative embodiments of the method may, for example, be employed to form the MIM capacitor according to the embodiments at FIGS. **6A-6C**, **12A**, and **12B**.

At **2302**, an interconnect structure is partially formed over a substrate, where the interconnect structure comprises a lower capacitor wire. See, for example, FIG. **14**.

At **2304**, a dielectric layer is formed over the lower capacitor wire. See, for example, FIG. **15**.

At **2306**, the dielectric layer is patterned to form a trench overlying and exposing the lower capacitor wire. See, for example, FIG. **16**.

At **2308**, a bottom electrode layer is deposited over the dielectric layer and lining the trench in a first process chamber, where a top surface of the bottom electrode layer has a first average surface roughness. See, for example, FIG. **17**. In alternative embodiments of the method, the patterning at **2306** is skipped so no trench is formed. Such alternative embodiments may, for example, be employed to form a MIM capacitor according to the embodiments at FIGS. **6A-6C**. In alternative embodiments of the method, the acts at **2302** and **2304** are skipped and the patterning at **2306** is performed to the substrate. Such alternative embodiments may, for example, be employed to form a MIM capacitor according to the embodiments at FIGS. **12A** and **12B**.

At **2310**, the bottom electrode layer is transferred to a second process chamber, where the transferring exposes the top surface of the bottom electrode layer to an oxidant and oxidizes the top surface, where the oxidizing forms a native oxide layer on the top surface, and where the top surface has a second average surface roughness greater than the first average surface roughness after the oxidizing. See, for example, FIG. **18**.

At **2312**, the native oxide layer and the top surface of the bottom electrode layer are plasma treated to smooth the top surface and form an interfacial layer in place of the native oxide layer, where the top surface has a third average surface roughness less than the second average surface roughness after the plasma treatment, and where the interfacial layer is conductive. See, for example, FIG. **19**. The plasma may, for example, be generated from a process gas being or comprising nitrous oxide and/or some other suitable process gas(es). In alternative embodiments of the method, the transferring at **2310** is skipped and the plasma treatment is performed in the first process chamber. See, for example, FIG. **19**.

At **2314**, an insulator layer is deposited on the interfacial layer, where the depositing of the insulator layer exposes the interfacial layer to an oxidant, and where the interfacial layer blocks the oxidant from diffusing to and oxidizing the top surface of the bottom electrode layer. See, for example, FIG. **20**. The insulator layer may, for example, be or comprise a metal oxide, a high k dielectric having a dielectric constant greater than about 10, some other suitable dielectric, or any combination of the foregoing.

At **2316**, a top electrode layer is deposited on the insulator layer. See, for example, FIG. **20**.

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At **2318**, the top electrode layer, the insulator layer, the interfacial layer, and the bottom electrode layer are patterned to form a MIM capacitor. See, for example, FIG. **21**. When forming the MIM capacitor according to embodiments in FIGS. **2A-2D**, the patterning may, for example, be performed by a photolithography/etching process or some other suitable patterning process. When forming the MIM capacitor according to embodiments in FIG. **2E**, the patterning may, for example, be performed by a planarization process to a top surface of the dielectric layer or by some other suitable patterning process.

At **2320**, the interconnect structure is completed around the MIM capacitor, where the interconnect structure comprises an upper capacitor electrode overlying the MIM capacitor and further comprises a TEVA extending from the upper capacitor electrode to the MIM capacitor. See, for example, FIG. **22**.

While the block diagram **2300** of FIG. **23** is illustrated and described herein as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events is not to be interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. Further, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein, and one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

In some embodiments, the present disclosure provides a method including: depositing a bottom electrode layer over a substrate; forming a native oxide layer on a top surface of the bottom electrode layer, wherein the native oxide layer has a first adhesion strength with the top surface of the bottom electrode layer; performing a plasma treatment process to replace the native oxide layer with an interfacial layer, wherein the interfacial layer is conductive and has a second adhesion strength with the top surface of the bottom electrode layer, and wherein the second adhesion strength is greater than the first adhesion strength; depositing an insulator layer on the interfacial layer; depositing a top electrode layer on the insulator layer; and patterning the top and bottom electrode layers, the insulator layer, and the interfacial layer to form a MIM capacitor. In some embodiments, the performing of the plasma treatment process reduces a thickness of the bottom electrode layer. In some embodiments, the bottom electrode layer includes titanium nitride, wherein the performing of the plasma treatment process includes: generating plasma from nitrogen oxide; and exposing the native oxide layer to the plasma, wherein nitrogen and oxygen diffuse into the native oxide layer and the bottom electrode layer to form a titanium oxynitride layer from the native oxide layer and a top portion of the bottom electrode layer. In some embodiments, the method further includes annealing the top and bottom electrode layers, the insulator layer, and the interfacial layer. In some embodiments, the method further includes patterning the substrate to form a trench, wherein the bottom electrode layer, the insulator layer, and the top electrode layer are deposited in the trench. In some embodiments, the method further includes: forming an alternating stack of wires and vias over the substrate; depositing a dielectric layer over the alternating stack; and patterning the dielectric layer to form a trench, wherein the bottom electrode layer, the insulator layer, and the top electrode layer are deposited in the trench.

In some embodiments, the present application provides another method including: depositing a bottom electrode layer over a substrate in a first process chamber; transferring

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the bottom electrode layer to a second process chamber, wherein a top surface of the bottom electrode layer has a first roughness upon completion of the transferring; plasma treating the top surface of the bottom electrode layer in the second process chamber, wherein the plasma treating forms an interfacial layer on the top surface of the bottom electrode layer, wherein the top surface of the bottom electrode layer has a second roughness less than the first roughness upon completion of the plasma treating; depositing an insulator layer on the interfacial layer; depositing a top electrode layer on the insulator layer; and patterning the top and bottom electrode layers, the insulator layer, and the interfacial layer to form a MIM capacitor. In some embodiments, the interfacial layer is exposed to an oxidant during the depositing of the insulator layer, wherein the interfacial layer blocks the oxidant from diffusing through the interfacial layer to the bottom electrode layer. In some embodiments, a thickness of the bottom electrode layer is a same before the depositing of the insulator layer as after the depositing of the insulator layer. In some embodiments, the bottom electrode layer includes titanium nitride, wherein the plasma treating forms plasma from nitrous oxide and exposes the top surface of the bottom electrode layer to the plasma. In some embodiments, the plasma treating includes generating plasma in the second process chamber and exposing the top surface of the bottom electrode layer to the plasma in the second process chamber, wherein the plasma is generated at a power of about 200-550 watts and at a pressure of about 1-2 torr, and wherein the top surface of the bottom electrode layer is exposed to the plasma for about 5-20 seconds. In some embodiments, the exposing of the top surface of the bottom electrode layer to the plasma is repeated multiple times. In some embodiments, the generating of the plasma includes introducing nitrous oxide into the second process chamber at a flow rate of about 10000-14000 SCCM.

In some embodiments, the present application provides a MIM capacitor including: a bottom electrode including a metal element and a non-metal element; an insulator layer overlying the bottom electrode; a top electrode overlying the insulator layer; and an interfacial layer between the bottom electrode and the insulator layer, wherein the interfacial layer directly contacts a top surface of the bottom electrode and is conductive, and wherein the interfacial layer includes the metal element, the non-metal element, and oxygen. In some embodiments, the interfacial layer is configured to block diffusion of oxygen from a top surface of the interfacial layer to the top surface of the bottom electrode. In some embodiments, the metal element includes titanium or tantalum, wherein the non-metal element includes nitrogen. In some embodiments, the interfacial layer includes titanium oxynitride, wherein an atomic percentage of oxygen in the interfacial layer decreases from top to bottom, and wherein an atomic percentage of nitrogen in the interfacial layer increases from top to bottom. In some embodiments, the interfacial layer includes titanium oxynitride, wherein an atomic percentage of oxygen in the interfacial layer is less than about 2 and greater than about 0, and wherein an atomic percentage of nitrogen in the interfacial layer is less than that of oxygen and greater than about 0. In some embodiments, the insulator layer includes a dielectric material having a dielectric constant greater than about 10. In some embodiments, the interfacial layer has a thickness of about 20-50 angstroms.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present

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disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A metal-insulator-metal (MIM) capacitor comprising:
a bottom electrode comprising a metal element and a non-metal element, wherein the non-metal element is different than oxygen;
a dielectric layer overlying the bottom electrode;
a top electrode overlying the dielectric layer; and
an interfacial layer between, and directly contacting, the bottom electrode and the dielectric layer, wherein the interfacial layer comprises oxide of the bottom electrode, the metal element, and the non-metal element; wherein an atomic percentage of oxygen in the interfacial layer decreases in a direction along a thickness of the interfacial layer, whereas an atomic percentage of the non-metal element in the interfacial layer increases in the direction along the thickness.
2. The MIM capacitor according to claim 1, wherein the atomic percentage of oxygen decreases from a top surface of the interfacial layer to a bottom surface of the interfacial layer, and wherein the atomic percentage of the non-metal element increases from the top surface of the interfacial layer to the bottom surface of the interfacial layer.
3. The MIM capacitor according to claim 1, wherein the bottom electrode, the interfacial layer, and the dielectric layer share a common width.
4. The MIM capacitor according to claim 1, wherein the interfacial layer is conductive.
5. The MIM capacitor according to claim 1, wherein the bottom electrode, the interfacial layer, and the dielectric layer have individual sidewalls arranged edge to edge to form a common sidewall.
6. The MIM capacitor according to claim 1, wherein the interfacial layer is titanium oxynitride, the metal element is titanium, and the non-metal element is nitrogen.
7. The MIM capacitor according to claim 1, wherein adhesion strength between the interfacial layer and the bottom electrode is greater than about 1000 micronewtons to prevent delamination.
8. A metal-insulator-metal (MIM) capacitor comprising:
a bottom electrode comprising a metal element and a non-metal element, wherein the non-metal element is different than oxygen;
a dielectric layer overlying the bottom electrode;
a top electrode overlying the dielectric layer; and
an interfacial layer between, and directly contacting, the bottom electrode and the dielectric layer, wherein the interfacial layer comprises oxide of the bottom electrode, the metal element, and the non-metal element; wherein the interfacial layer has a pair of first sidewalls respectively on opposite sides of the MIM capacitor and extending from a top surface of the interfacial layer, and wherein the bottom electrode has a pair of

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- second sidewalls respectively on the opposite sides and separated by substantially a same distance as the first sidewalls.
9. The MIM capacitor according to claim 8, wherein the dielectric layer is between and directly contacts the top electrode and the interfacial layer.
 10. The MIM capacitor according to claim 8, wherein an atomic percentage of oxygen in the interfacial layer is greater at a bottom surface of the interfacial layer than at a top surface of the interfacial layer.
 11. The MIM capacitor according to claim 8, wherein an atomic percentage of the non-metal element in the interfacial layer is greater at a bottom surface of the interfacial layer than at the top surface of the interfacial layer.
 12. The MIM capacitor according to claim 8, wherein a thickness of the interfacial layer is about 20-50 angstroms.
 13. The MIM capacitor according to claim 8, wherein the second sidewalls are respectively edge to edge with the first sidewalls.
 14. The MIM capacitor according to claim 8, wherein the bottom electrode, the interfacial layer, and the dielectric layer have individual U-shaped profiles, and wherein top surfaces respectively of the U-shaped profiles are level with each other and a top surface of the top electrode.
 15. An integrated circuit (IC) chip comprising a metal-insulator-metal (MIM) capacitor, wherein the MIM capacitor comprises:
a bottom electrode comprising a metal nitride;
a metal oxide layer overlying the bottom electrode;
a top electrode overlying and directly contacting the metal oxide layer; and
a metal oxynitride layer between, and directly contacting, the bottom electrode and the metal oxide layer, wherein the metal oxynitride layer comprises the metal nitride and has a higher atomic percentage of oxygen than nitrogen at a bottom surface of the metal oxynitride layer.
 16. The IC chip according to claim 15, wherein the metal oxynitride layer has the higher atomic percentage of oxygen than nitrogen at a top surface of the metal oxynitride layer.
 17. The IC chip according to claim 15, wherein the metal oxynitride layer is TiO_xN_y , wherein x and y are numbers, wherein $x > y$, and wherein x is less than 2 and greater than 0.
 18. The IC chip according to claim 15, wherein a top surface of the bottom electrode and the top surface of the metal oxynitride layer share a common width.
 19. The IC chip according to claim 15, further comprising:
a first wire and a second wire underlying the MIM capacitor at a first common elevation;
a third wire and a fourth wire overlying the MIM capacitor at a second common elevation;
a first via and a second via extending from the MIM capacitor respectively to the first and third wires; and
a third via extending from the second wire to the fourth wire.
 20. The IC chip according to claim 19, further comprising:
a semiconductor substrate; and
a transistor on the semiconductor substrate, wherein the first wire and the second wire are spaced over the transistor, and wherein the first wire is electrically coupled to the transistor.

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